



Group Assessment Coversheet

To be attached to the front of the assessment.

Campus: Bedfordview

Faculty: Information Technology

Module Code: ITNSA2-44

Group: 1

Lecturer's Name: Mr Tawanda Dundhlu

Indicate	Yes	No
Plagiarism report attached		

Declaration:

I declare that this assessment is my own original work except for source material explicitly acknowledged. I also declare that this assessment or any other of my original work related to it has not been previously, or is not being simultaneously, submitted for this or any other course. I am aware of the AI policy and acknowledge that I have not used any AI technology to generate or manipulate data, other than as permitted by the assessment instructions. I also declare that I am aware of the Institution's policy and regulations on honesty in academic work as set out in the Conditions of Enrolment, and of the disciplinary guidelines applicable to breaches of such policy and regulations.

Eduvos (Pty) Ltd. (formerly Pearson Institute of Higher Education) is registered with the
Department of Higher Education and institution under the Higher Education Act, 101, of 1997.

Registration Certificate number: 2001/HE07/008

Training as a private higher education

			% Participated
1	Student Full Name	Qhawe Nze	
	Student Number	EDUV4956354	
	Contact Number		
	Signature		
2	Student Full Name	Sihle Ngwenya	
	Student Number	EDUV4948242	
	Contact Number		
	Signature		
3	Student Full Name	Eliandro Miranda	
	Student Number	EDUV4966829	
	Contact Number		
	Signature		
4	Student Full Name	Kresan Padayachee	
	Student Number	EDUV4965243	
	Contact Number		
	Signature		

Lecturer's Comments:

--

Marks Awarded:

%

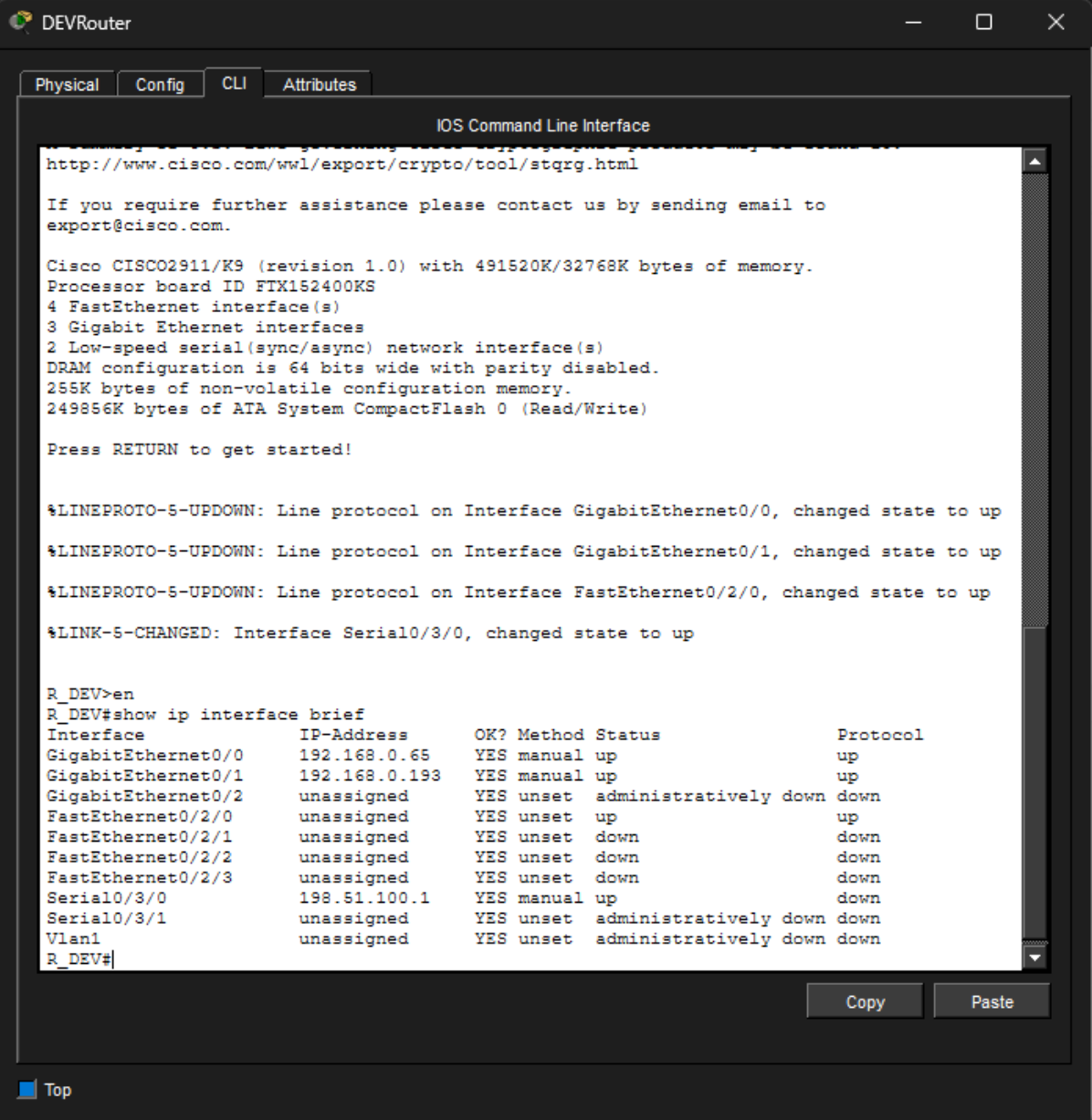
Signature

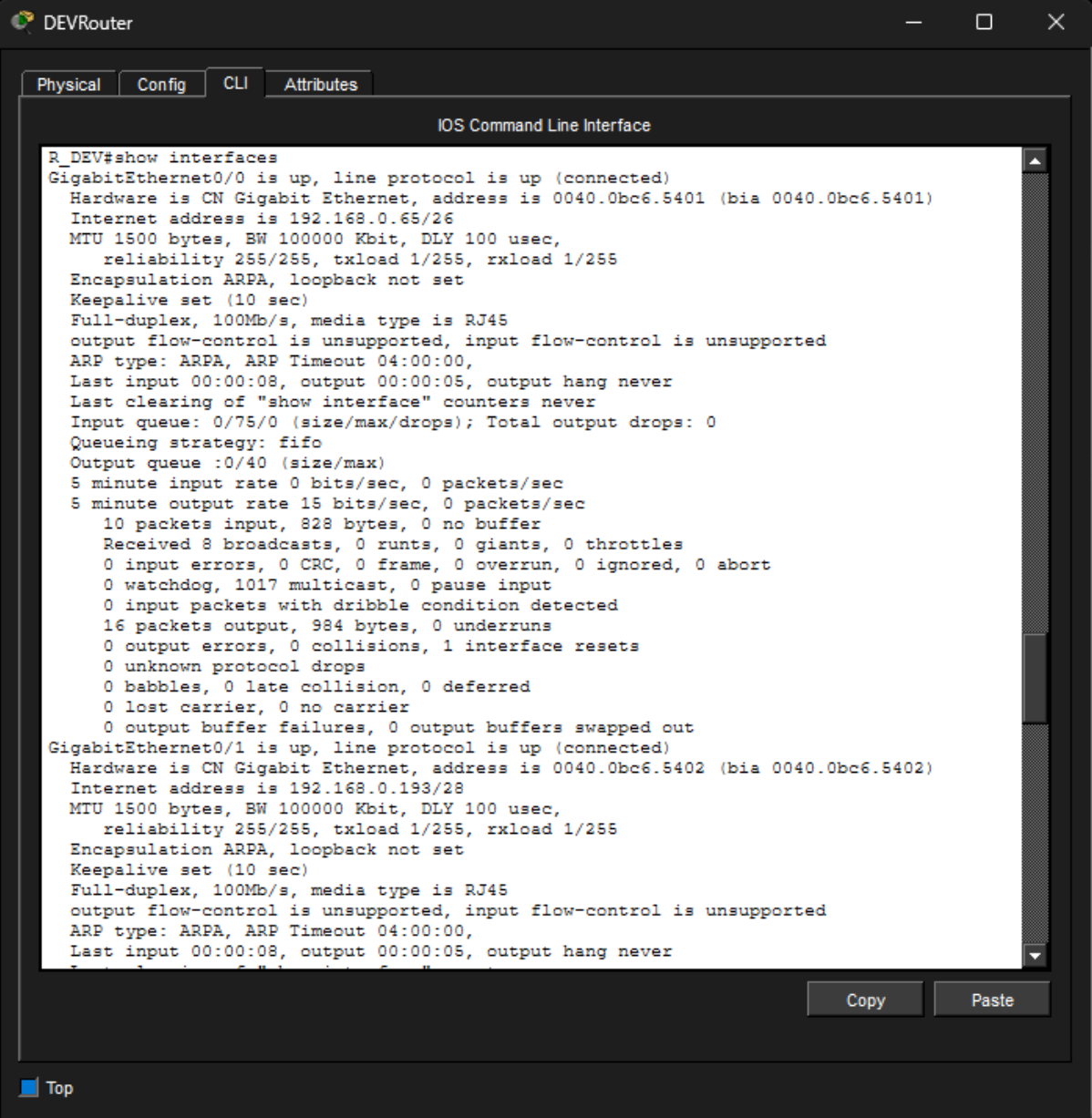
Date

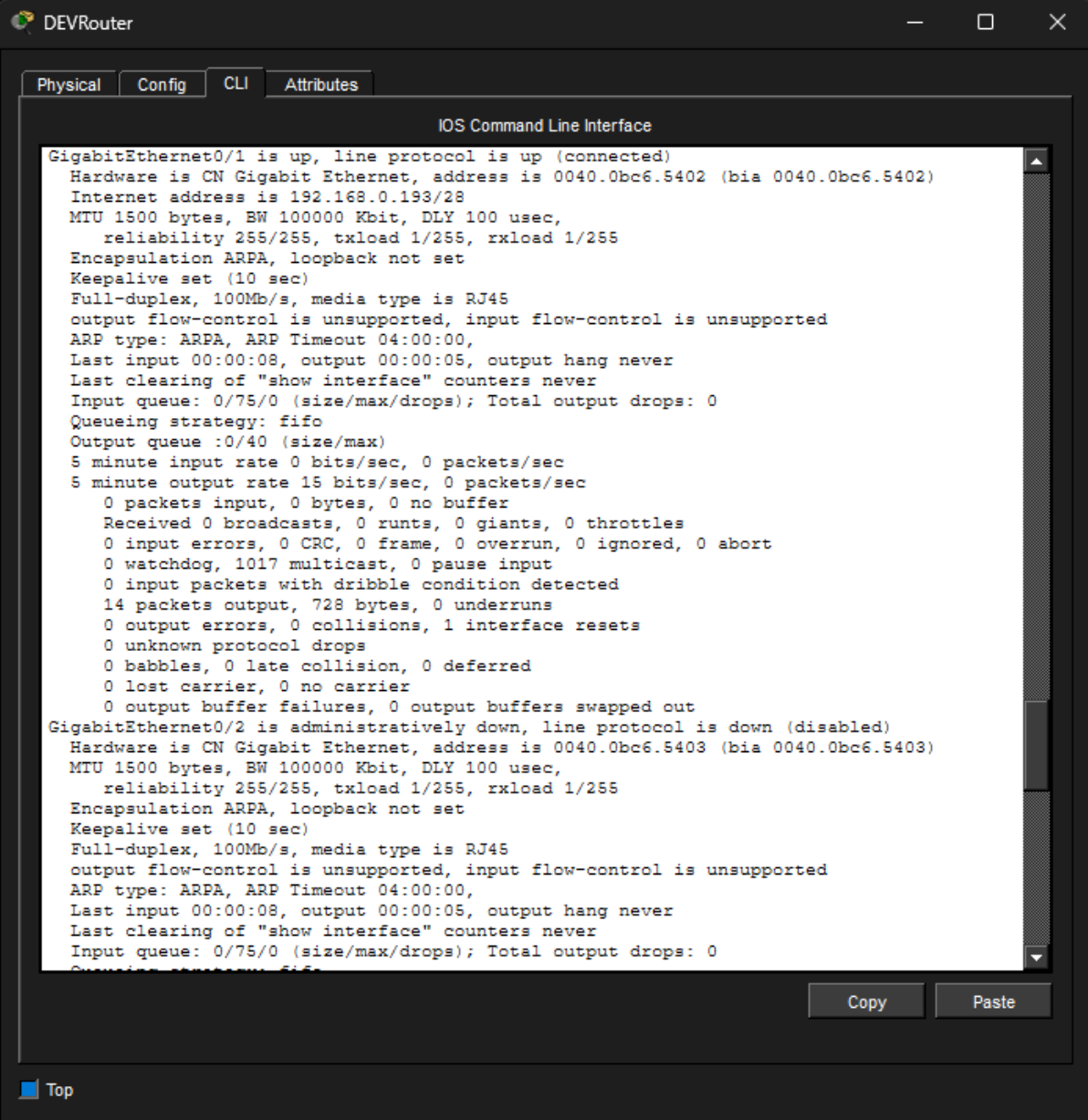
Question 1

1.1

Dev Router





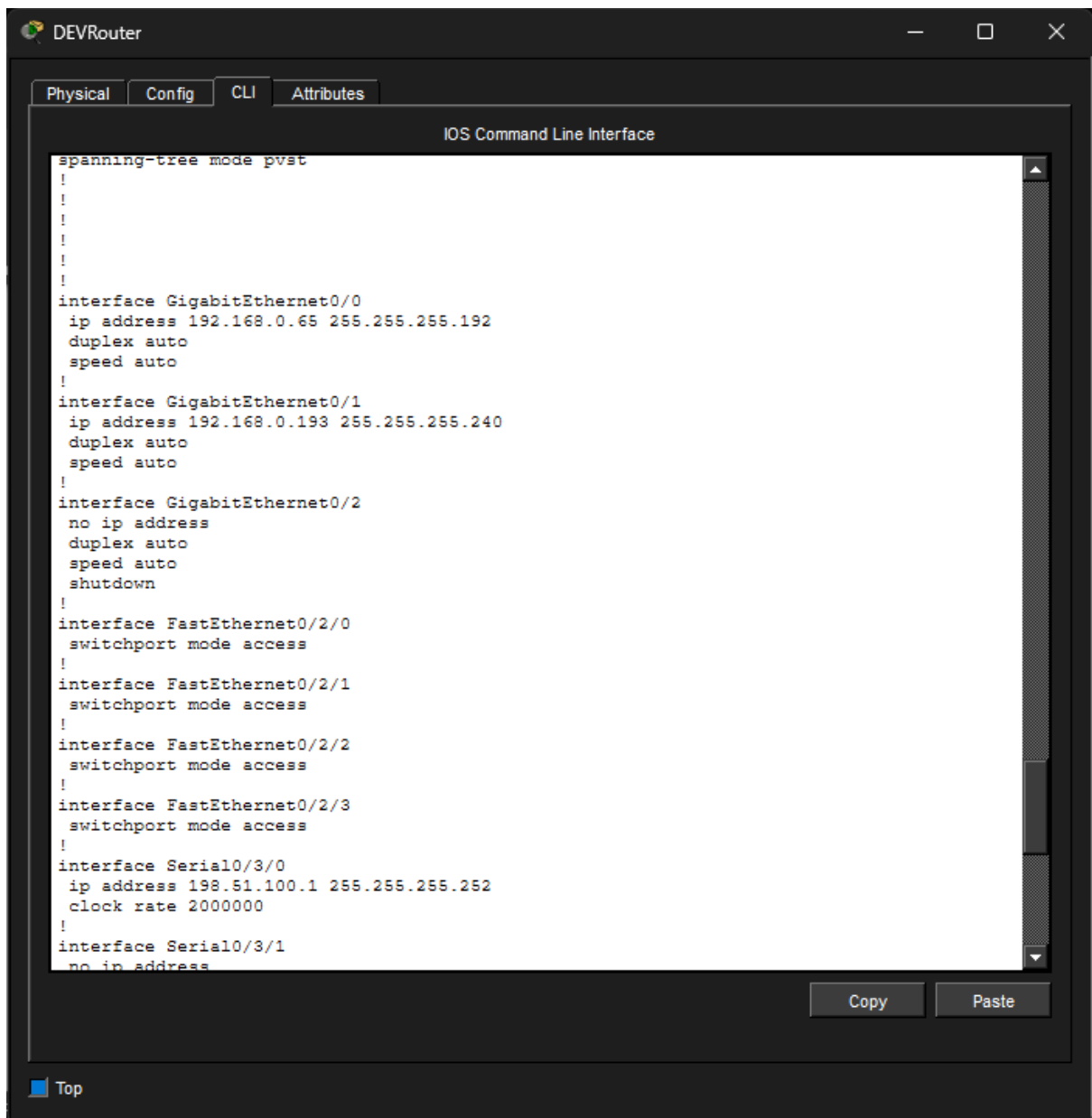


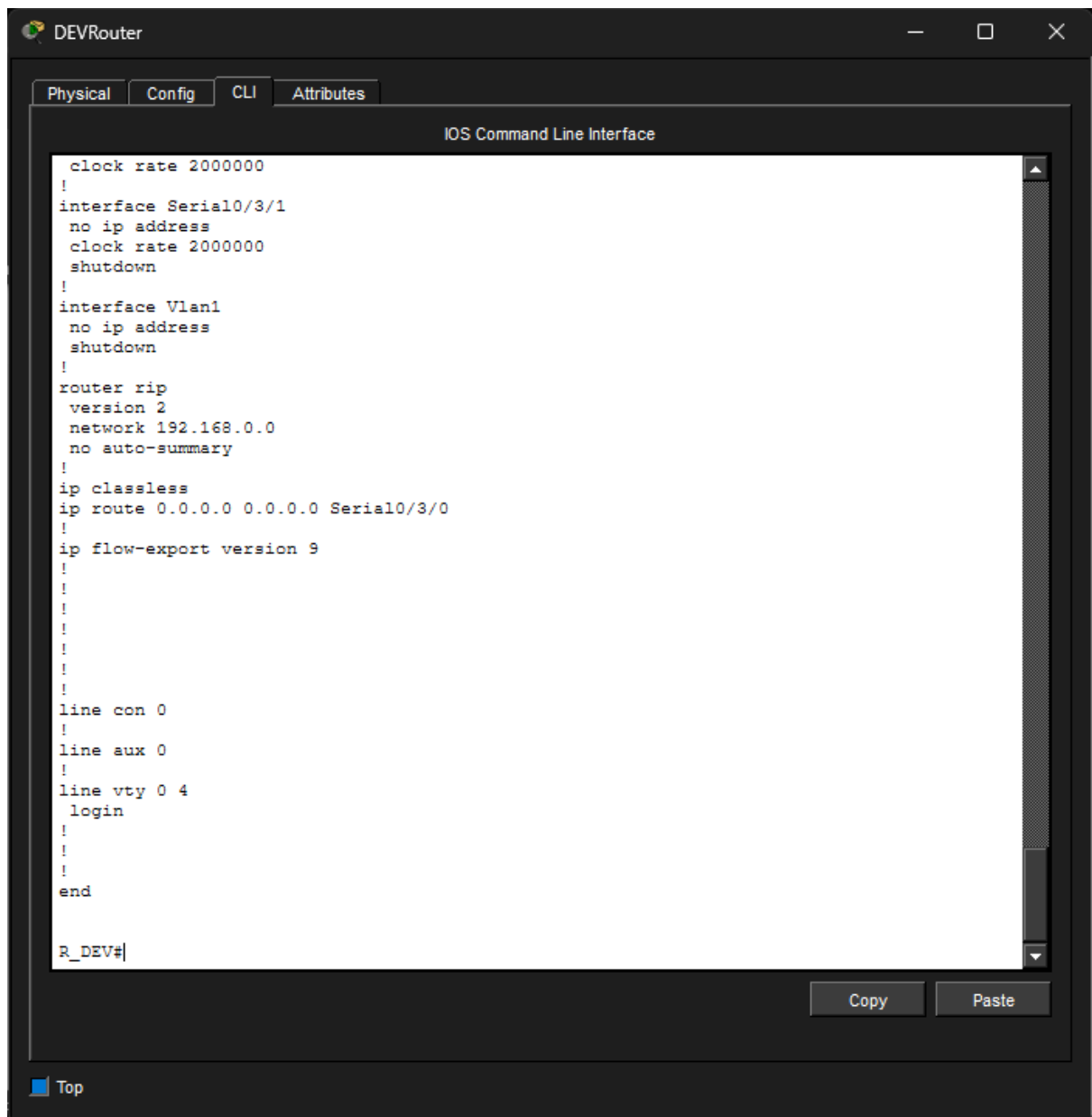
IOS Command Line Interface

```
Serial0/3/0 is up, line protocol is down (disabled)
Hardware is HD64570
Internet address is 198.51.100.1/30
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation HDLC, loopback not set, keepalive set (10 sec)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
    Conversations 0/0/256 (active/max active/max total)
    Reserved Conversations 0/0 (allocated/max allocated)
    Available Bandwidth 1158 kilobits/sec
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 output buffer failures, 0 output buffers swapped out
    0 carrier transitions
DCD=up DSR=up DTR=up RTS=up CTS=up
Serial0/3/1 is administratively down, line protocol is down (disabled)
Hardware is HD64570
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation HDLC, loopback not set, keepalive set (10 sec)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
    Conversations 0/0/256 (active/max active/max total)
    Reserved Conversations 0/0 (allocated/max allocated)
    Available Bandwidth 1158 kilobits/sec
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
```

Copy

Paste





DEV Switch

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
```

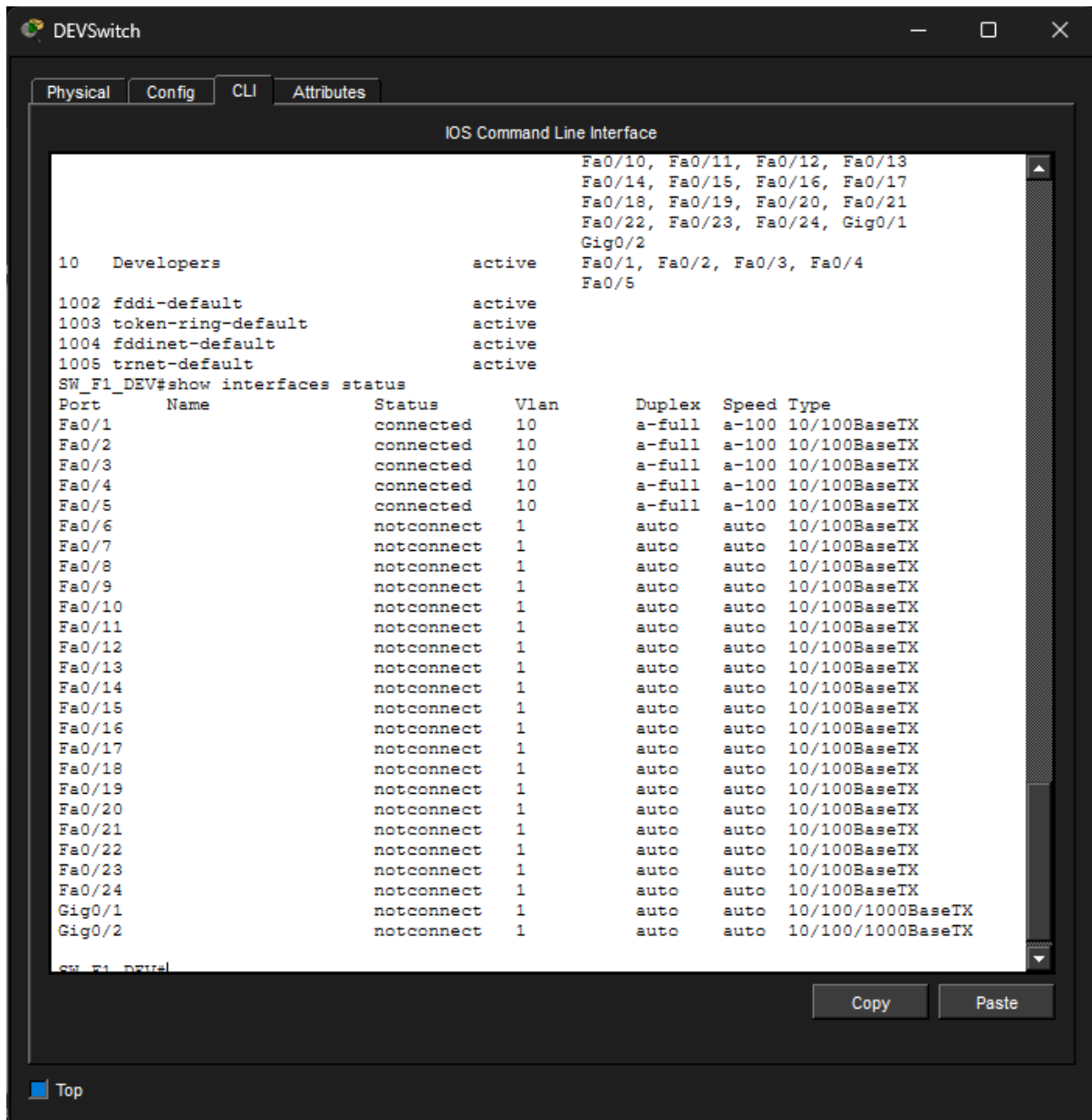
```
SW_F1_DEV>en
SW_F1_DEV#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10 Developers	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
SW_F1_DEV#
```

Copy

Paste



DEV PCs and Server

DEVPC1

Physical

Config

Desktop

Programming

Attributes

IP Configuration

Interface

FastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

192.168.0.68

Subnet Mask

255.255.255.192

Default Gateway

0.0.0.0

DNS Server

192.168.0.13

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::205:5EFF:FE14:BE1D

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

Top

DEVServer

Physical

Config

Services

Desktop

Programming

Attributes

IP Configuration

IP Configuration

DHCP

Static

IPv4 Address

192.168.0.70

Subnet Mask

255.255.255.192

Default Gateway

192.168.0.65

DNS Server

192.168.0.13

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::260:2FFF:FE7E:352A

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

Top

BA Router

BARouter

Physical

Config

CLI

Attributes

IOS Command Line Interface

```
Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
4 FastEthernet interface(s)
3 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2/0, changed state to up
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up

R_BA>en
R_BA#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0       192.168.0.129   YES manual    up              up
GigabitEthernet0/1       unassigned      YES unset     administratively down down
GigabitEthernet0/2       192.168.0.213   YES manual    up              up
FastEthernet0/2/0        unassigned      YES unset     up              up
FastEthernet0/2/1        unassigned      YES unset     down            down
FastEthernet0/2/2        unassigned      YES unset     down            down
FastEthernet0/2/3        unassigned      YES unset     down            down
Serial0/3/0              198.51.100.5    YES manual    up              down
Serial0/3/1              192.168.2.1     YES manual    up              up
Vlan1                    unassigned      YES unset     administratively down down
R_BA#
```

Copy

Paste

Top

IOS Command Line Interface

```
R_BA#show interfaces
GigabitEthernet0/0 is up, line protocol is up (connected)
  Hardware is CN Gigabit Ethernet, address is 00d0.d316.bb01 (bia 00d0.d316.bb01)
  Internet address is 192.168.0.129/26
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s, media type is RJ45
  output flow-control is unsupported, input flow-control is unsupported
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 21 bits/sec, 0 packets/sec
    10 packets input, 828 bytes, 0 no buffer
      Received 8 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 1017 multicast, 0 pause input
    0 input packets with dribble condition detected
    36 packets output, 2704 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
GigabitEthernet0/1 is administratively down, line protocol is down (disabled)
  Hardware is CN Gigabit Ethernet, address is 00d0.d316.bb02 (bia 00d0.d316.bb02)
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s, media type is RJ45
  output flow-control is unsupported, input flow-control is unsupported
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
```

Copy

Paste

IOS Command Line Interface

```
GigabitEthernet0/2 is up, line protocol is up (connected)
Hardware is CN Gigabit Ethernet, address is 00d0.d316.bb03 (bia 00d0.d316.bb03)
Internet address is 192.168.0.213/30
MTU 1500 bytes, BW 1000000 Kbit, DLY 10 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s, media type is RJ45
output flow-control is unsupported, input flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00,
Last input 00:00:08, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: fifo
Output queue :0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 21 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 1017 multicast, 0 pause input
    0 input packets with dribble condition detected
    34 packets output, 2448 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
FastEthernet0/2/0 is up, line protocol is up (connected)
Hardware is Lance, address is 000a.f34b.9401 (bia 000a.f34b.9401)
BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s
input flow-control is off, output flow-control is off
ARP type: ARPA, ARP Timeout 00:04:00
Last input 00:00:08, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
```

Copy

Paste

IOS Command Line Interface

```
FastEthernet0/2/0 is up, line protocol is up (connected)
  Hardware is Lance, address is 000a.f34b.9401 (bia 000a.f34b.9401)
  BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    956 packets input, 193351 bytes, 0 no buffer
    Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 0 multicast, 0 pause input
    0 input packets with dribble condition detected
    2357 packets output, 263570 bytes, 0 underruns
    0 output errors, 0 collisions, 10 interface resets
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
FastEthernet0/2/1 is up, line protocol is down (disabled)
  Hardware is Lance, address is 000a.f34b.9402 (bia 000a.f34b.9402)
  BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
```

Copy

Paste

IOS Command Line Interface

```
Serial0/3/0 is up, line protocol is down (disabled)
Hardware is HD64570
Internet address is 198.51.100.5/30
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation HDLC, loopback not set, keepalive set (10 sec)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
    Conversations 0/0/256 (active/max active/max total)
    Reserved Conversations 0/0 (allocated/max allocated)
    Available Bandwidth 1158 kilobits/sec
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 output buffer failures, 0 output buffers swapped out
    0 carrier transitions
DCD=up DSR=up DTR=up RTS=up CTS=up
Serial0/3/1 is up, line protocol is up (connected)
Hardware is HD64570
Internet address is 192.168.2.1/30
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation HDLC, loopback not set, keepalive set (10 sec)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
    Conversations 0/0/256 (active/max active/max total)
    Reserved Conversations 0/0 (allocated/max allocated)
    Available Bandwidth 1158 kilobits/sec
5 minute input rate 51 bits/sec, 0 packets/sec
5 minute output rate 21 bits/sec, 0 packets/sec
```

Copy

Paste

IOS Command Line Interface

```
Serial0/3/1 is up, line protocol is up (connected)
Hardware is HD64570
Internet address is 192.168.2.1/30
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation HDLC, loopback not set, keepalive set (10 sec)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
    Conversations 0/0/256 (active/max active/max total)
    Reserved Conversations 0/0 (allocated/max allocated)
    Available Bandwidth 1158 kilobits/sec
5 minute input rate 51 bits/sec, 0 packets/sec
5 minute output rate 21 bits/sec, 0 packets/sec
    95 packets input, 6080 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    35 packets output, 2500 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 output buffer failures, 0 output buffers swapped out
    0 carrier transitions
DCD=up DSR=up DTR=up RTS=up CTS=up
Vlan1 is administratively down, line protocol is down
Hardware is CPU Interface, address is 0001.432b.ac60 (bia 0001.432b.ac60)
MTU 1500 bytes, BW 100000 Kbit, DLY 1000000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
ARP type: ARPA, ARP Timeout 04:00:00
Last input 21:40:21, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
    1682 packets input, 530955 bytes, 0 no buffer
    Received 0 broadcasts (0 IP multicast)
    0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
```

Copy

Paste

IOS Command Line Interface

```
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: weighted fair
Output queue: 0/1000/64/0 (size/max total/threshold/drops)
  Conversations 0/0/256 (active/max active/max total)
  Reserved Conversations 0/0 (allocated/max allocated)
  Available Bandwidth 1158 kilobits/sec
5 minute input rate 51 bits/sec, 0 packets/sec
5 minute output rate 21 bits/sec, 0 packets/sec
  95 packets input, 6080 bytes, 0 no buffer
  Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
  35 packets output, 2500 bytes, 0 underruns
  0 output errors, 0 collisions, 0 interface resets
  0 output buffer failures, 0 output buffers swapped out
  0 carrier transitions
DCD=up DSR=up DTR=up RTS=up CTS=up
Vlan1 is administratively down, line protocol is down
Hardware is CPU Interface, address is 0001.432b.ac60 (bia 0001.432b.ac60)
MTU 1500 bytes, BW 1000000 Kbit, DLY 10000000 usec,
  reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
ARP type: ARPA, ARP Timeout 04:00:00
Last input 21:40:21, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
  1682 packets input, 530955 bytes, 0 no buffer
  Received 0 broadcasts (0 IP multicast)
  0 runts, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
  563859 packets output, 0 bytes, 0 underruns
  0 output errors, 23 interface resets
  0 output buffer failures, 0 output buffers swapped out
```

R_BA#

Copy

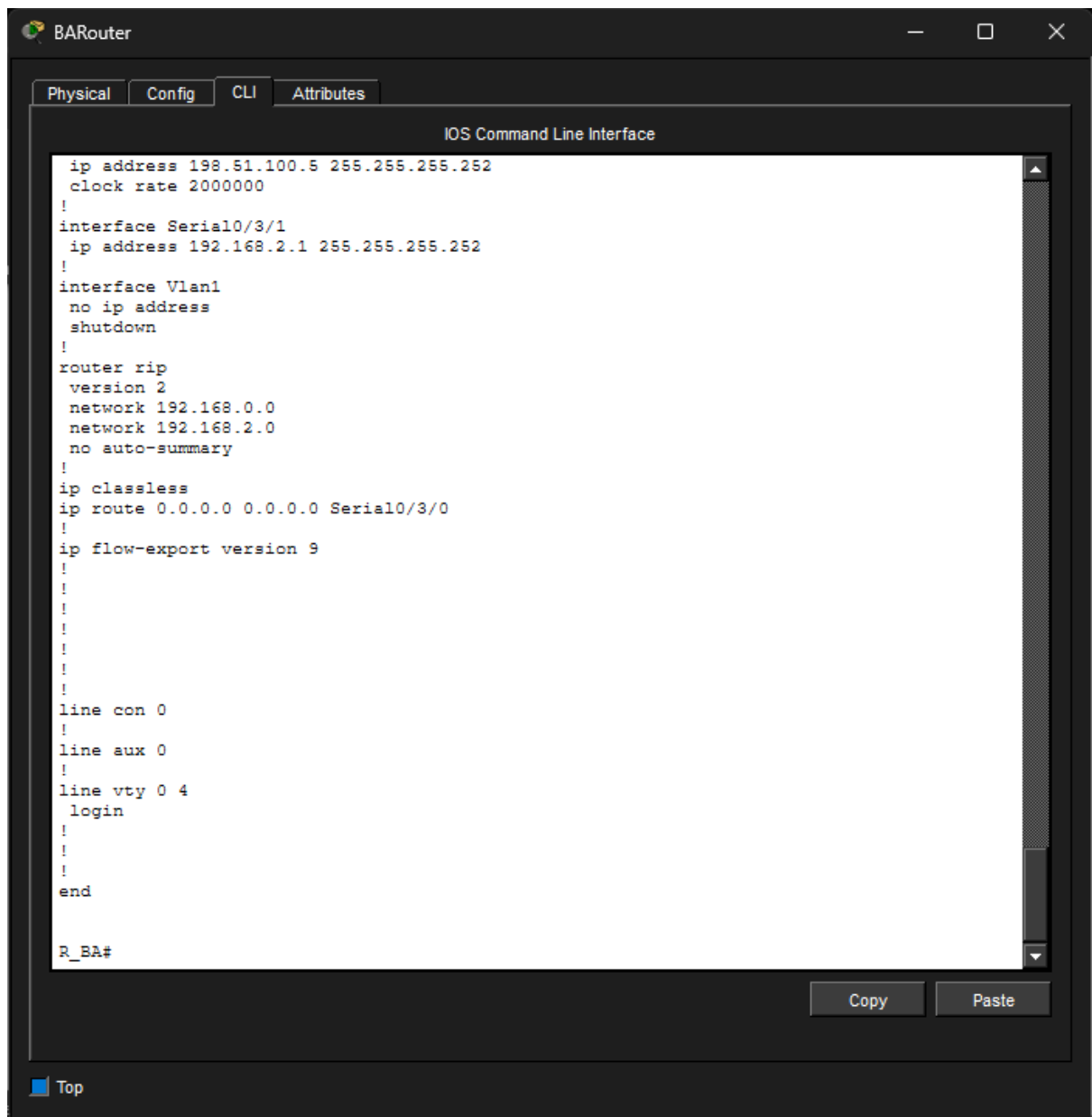
Paste

IOS Command Line Interface

```
spanning-tree mode pvst
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
ip address 192.168.0.129 255.255.255.192  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/1  
no ip address  
duplex auto  
speed auto  
shutdown  
!  
interface GigabitEthernet0/2  
ip address 192.168.0.213 255.255.255.252  
duplex auto  
speed auto  
!  
interface FastEthernet0/2/0  
switchport mode access  
!  
interface FastEthernet0/2/1  
switchport mode access  
!  
interface FastEthernet0/2/2  
switchport mode access  
!  
interface FastEthernet0/2/3  
switchport mode access  
!  
interface Serial0/3/0  
ip address 198.51.100.5 255.255.255.252  
clock rate 2000000  
!  
interface Serial0/3/1  
ip address 192.168.0.1 255.255.255.252
```

Copy

Paste



BA Switch

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
```

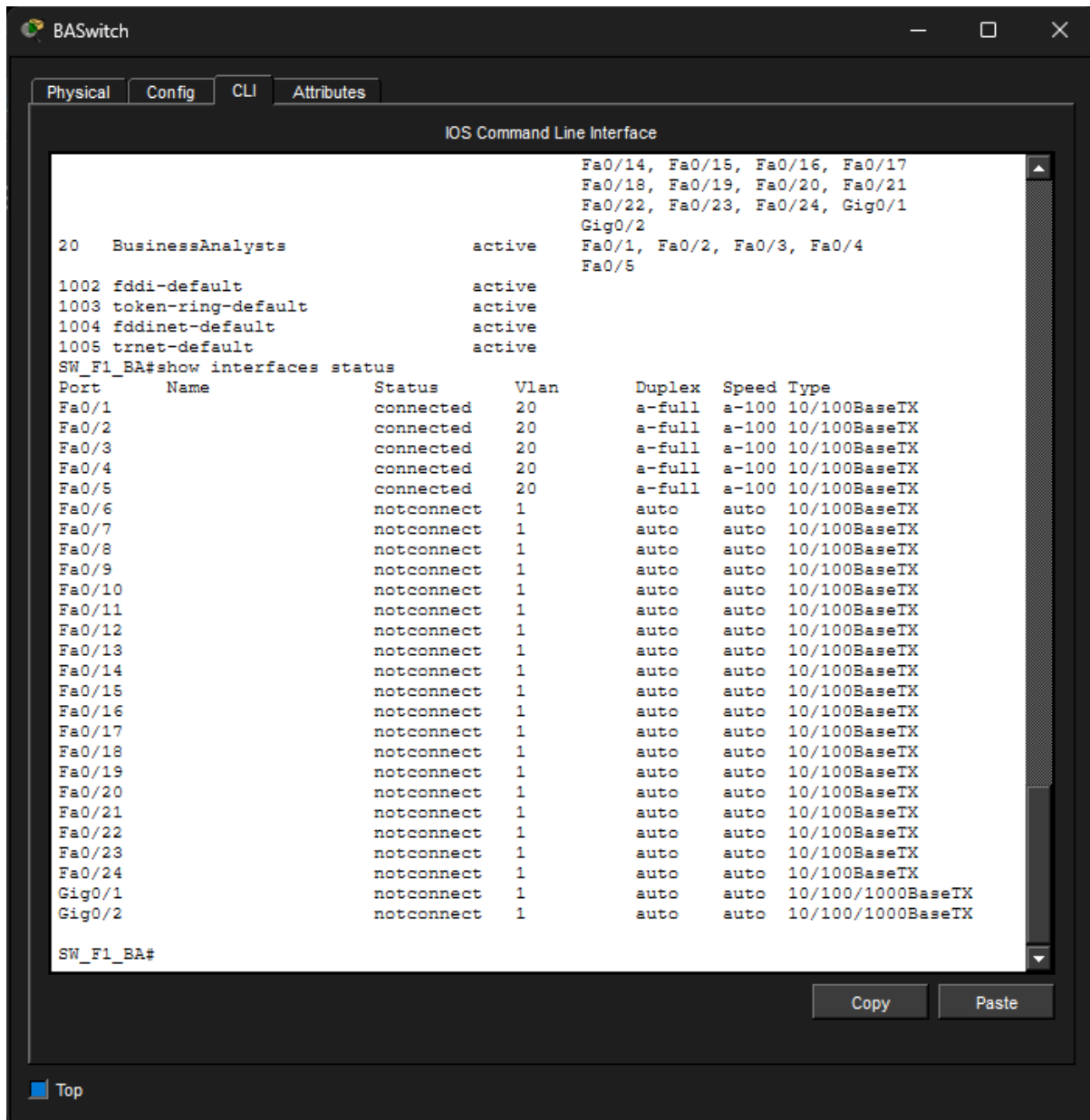
```
SW_F1_BA>en
SW_F1_BA#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
20 BusinessAnalysts	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
SW_F1_BA#
```

Copy

Paste



BA PCs and Server

IP Configuration

X

Interface FastEthernet0

IP Configuration

☒ DHCP☒ Static

IPv4 Address

192.168.0.132

Subnet Mask

255.255.255.192

Default Gateway

0.0.0.0

DNS Server

192.168.0.13

IPv6 Configuration

☒ Automatic☒ Static

IPv6 Address

Link Local Address

FE80::202:16FF:FEC8:90AE

Default Gateway

DNS Server

802.1X

☒ Use 802.1X Security

Authentication

MD5

Username

Password

BAServer

Physical

Config

Services

Desktop

Programming

Attributes

IP Configuration

DHCP

Static

IPv4 Address

192.168.0.134

Subnet Mask

255.255.255.192

Default Gateway

192.168.0.129

DNS Server

192.168.0.13

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::260:2FFF:FE74:E4D9

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

Top

DEV Server

DEVServer

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

DHCP

Interface

FastEthernet0

Service

On

Off

Pool Name

DEV Pool

Default Gateway

192.168.0.65

DNS Server

192.168.0.13

Start IP Address :

192

168

0

66

Subnet Mask:

255

255

255

192

Maximum Number of Users :

50

TFTP Server:

0.0.0.0

WLC Address:

0.0.0.0

Add

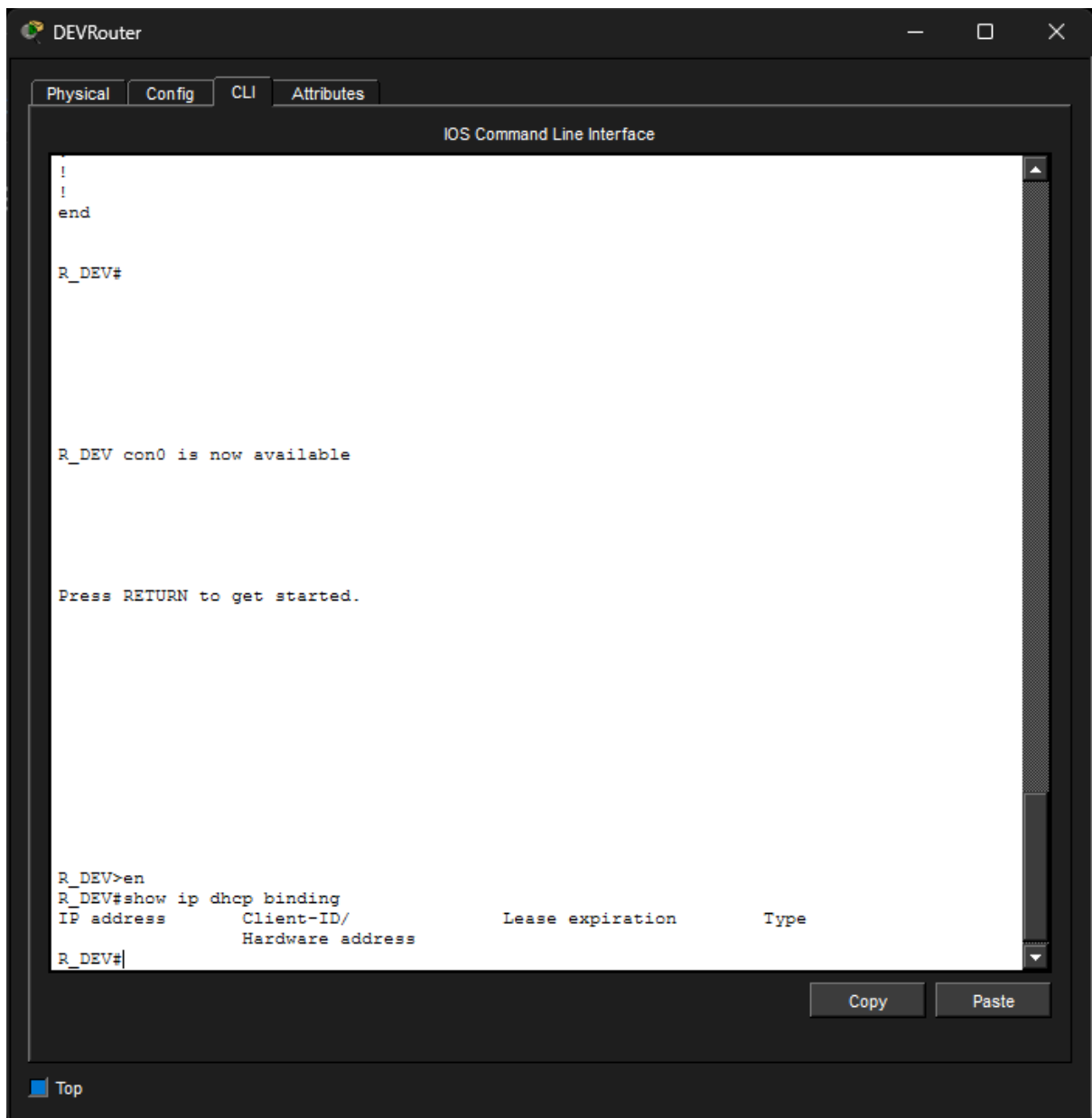
Save

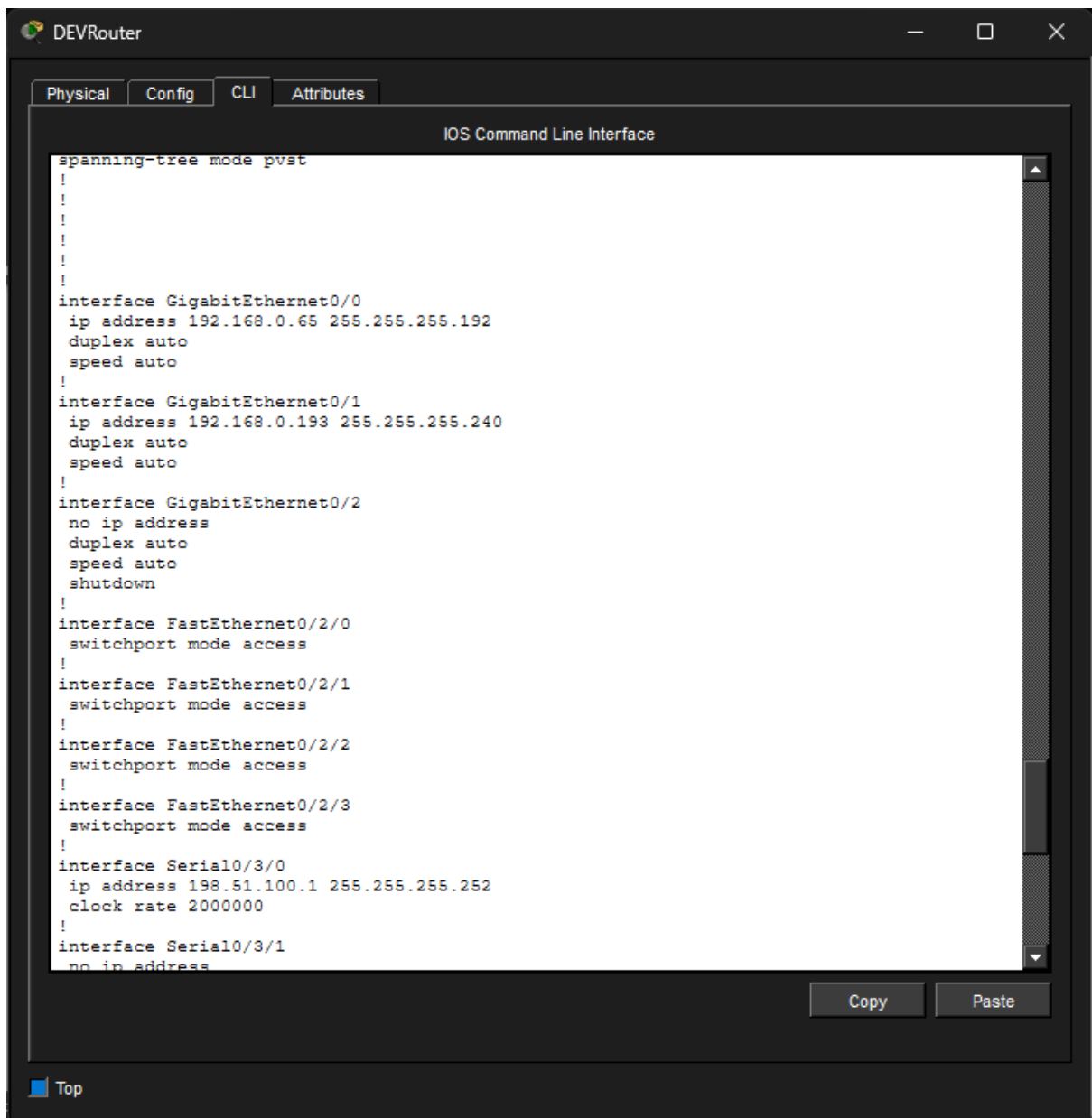
Remove

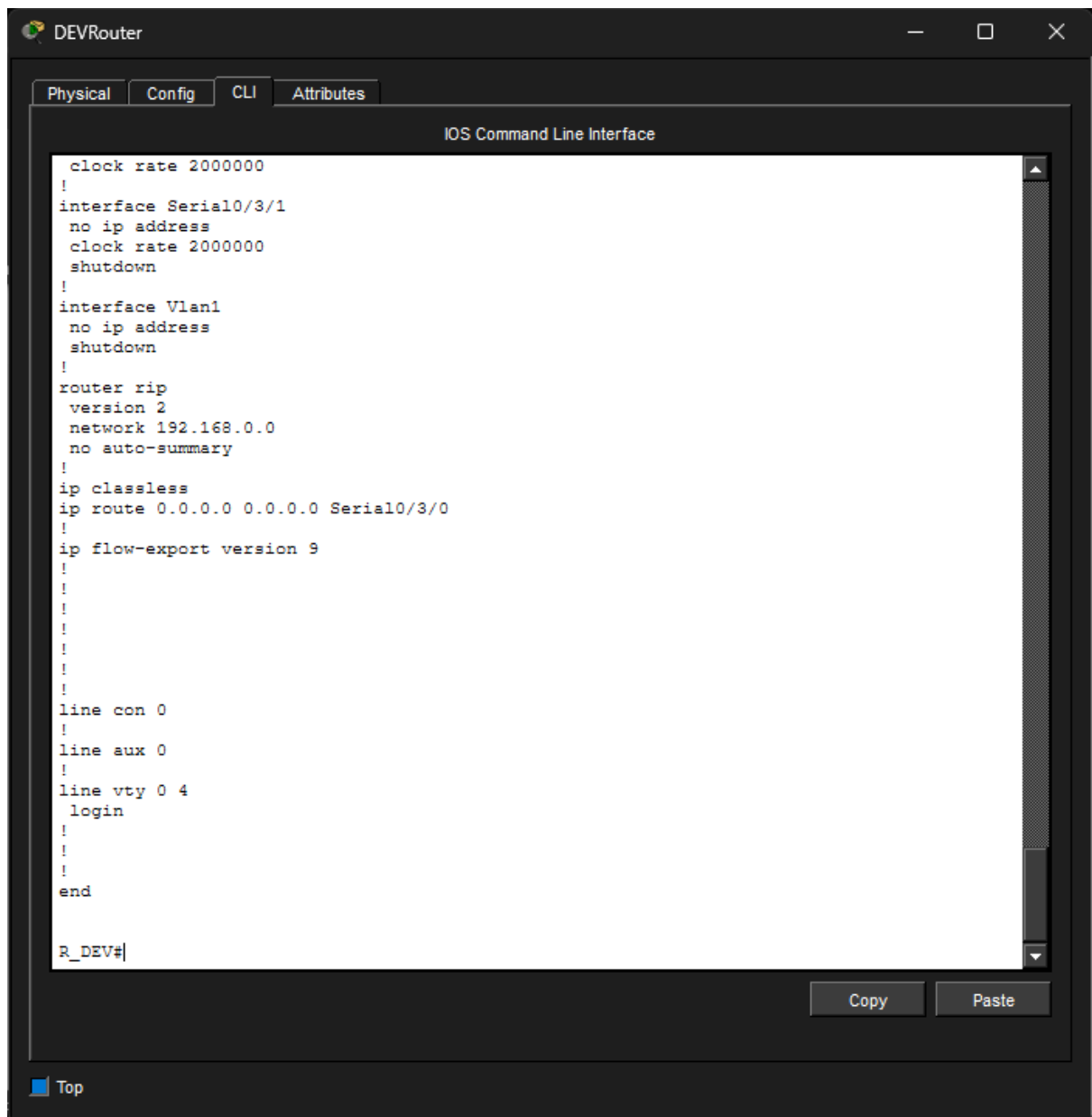
Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max Uset	TFTP Server	WLC Address
DEV Pool	192.168....	192.168....	192.168....	255.255....	50	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168....	255.255....	512	0.0.0.0	0.0.0.0

Top

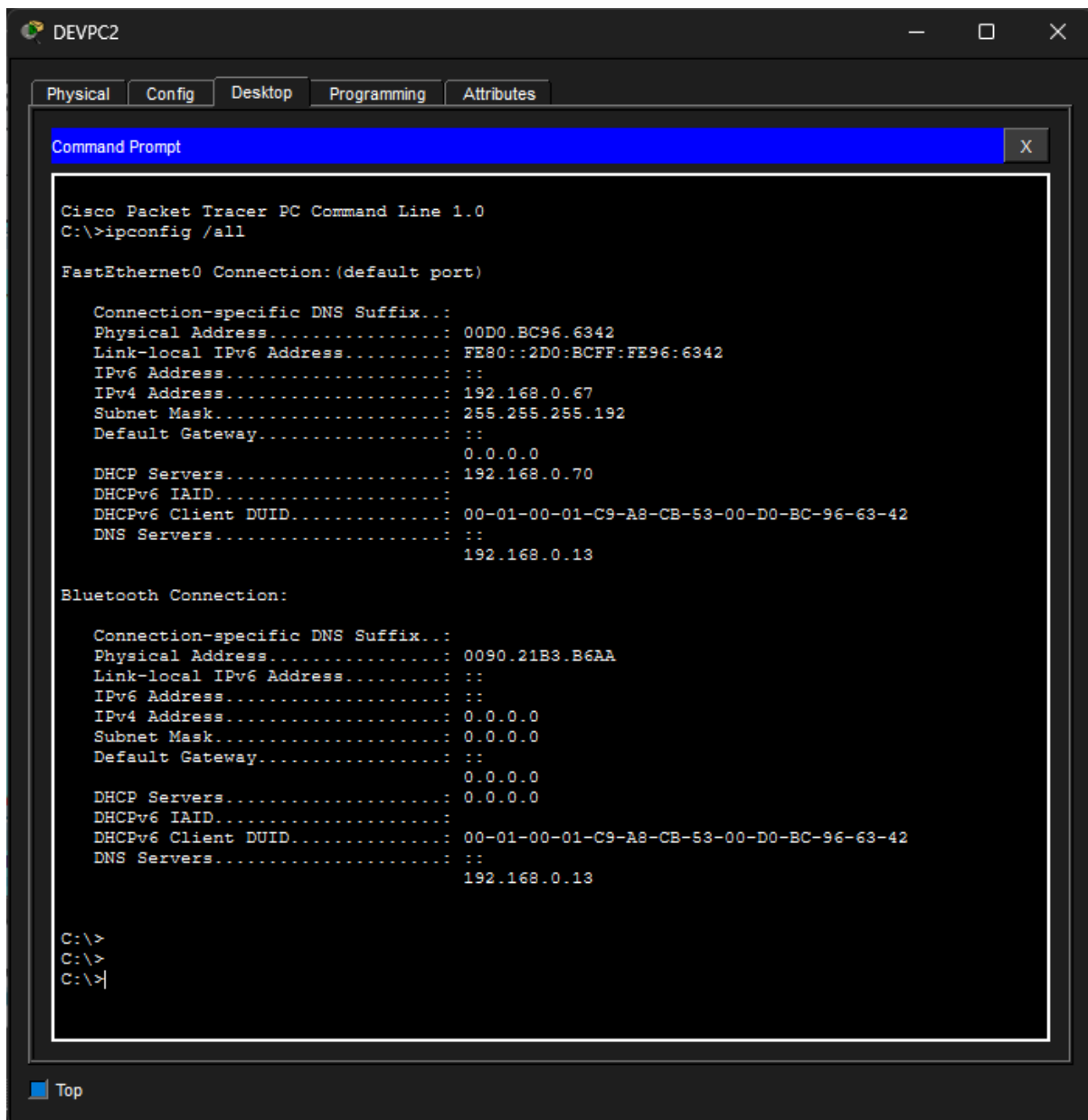
DEV Router







Client Verification



BA Server

BA Server

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

DHCP

InterfaceFastEthernet0ServiceOnOff

Pool NameBA Pool

Default Gateway192.168.0.129

DNS Server192.168.0.13

Start IP Address :1921680130

Subnet Mask:255255255192

Maximum Number of Users :50

TFTP Server:0.0.0.0

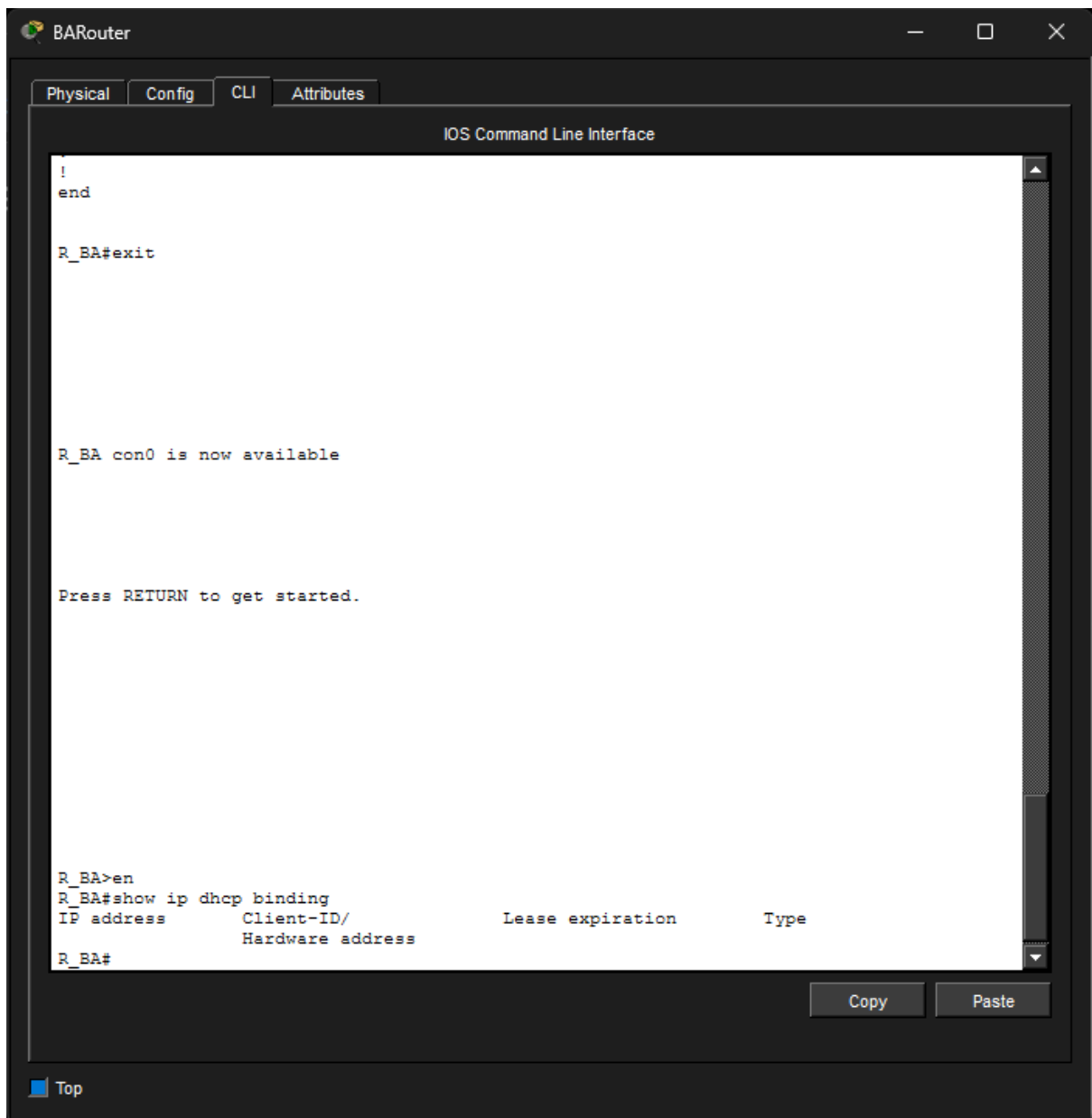
WLC Address:0.0.0.0

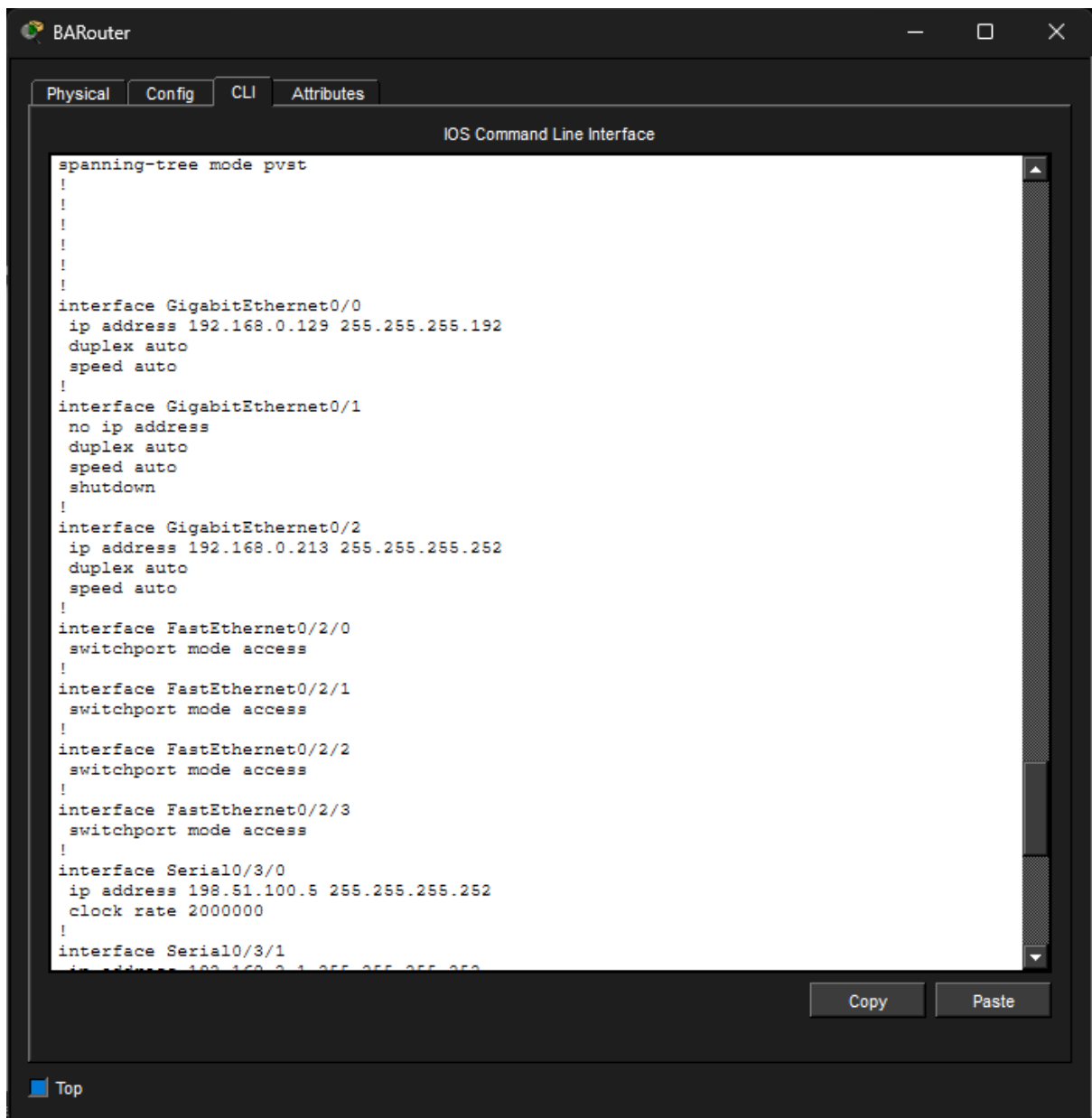
AddSaveRemove

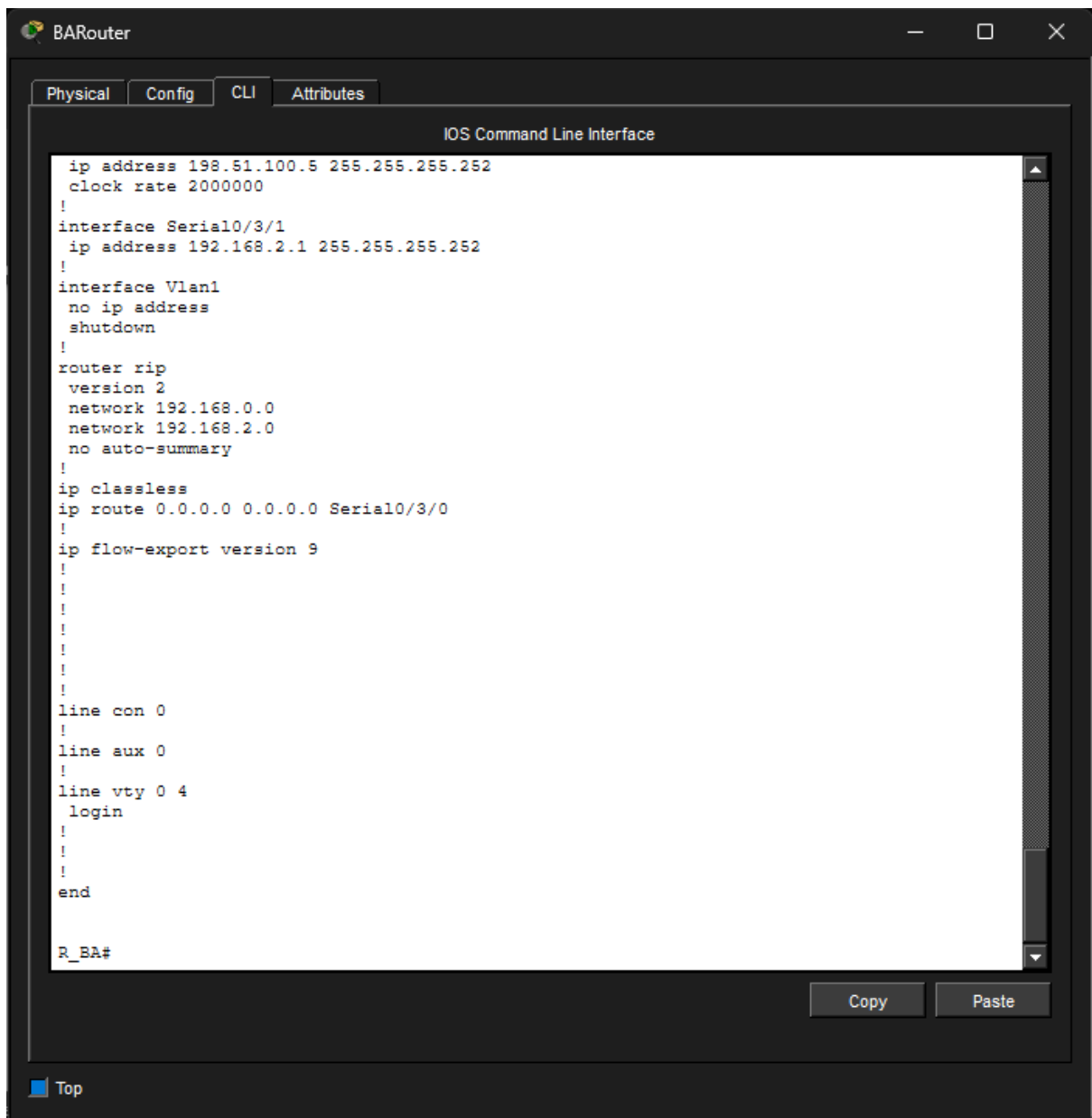
Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
BA Pool	192.168....	192.168....	192.168....	255.255....	50	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168....	255.255....	512	0.0.0.0	0.0.0.0

Top

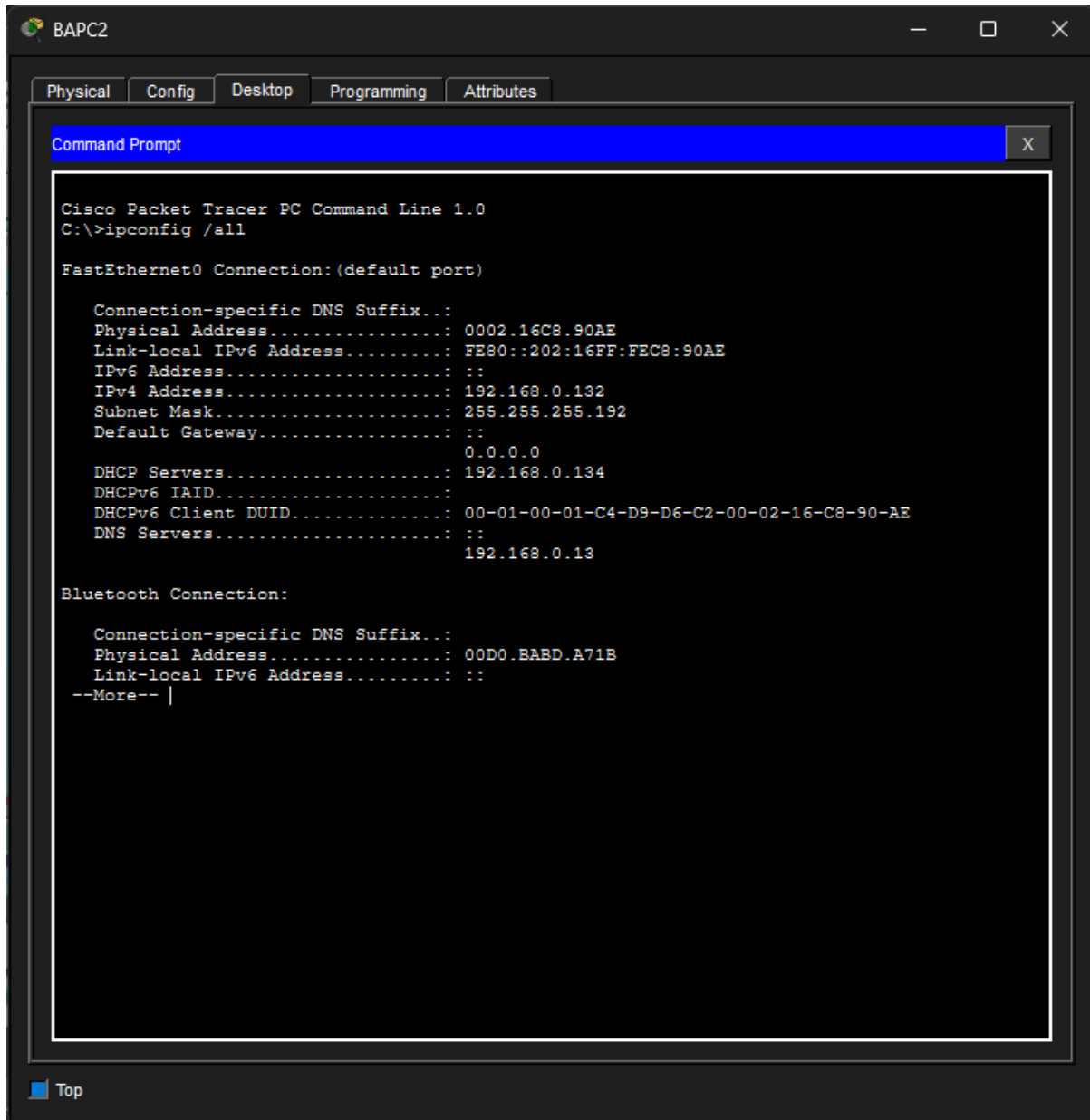
BA Router







Client Verification



1.3

Central DNS Server records

Central DNS Server

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

DNS

DNS Service

On

Off

Resource Records

Name

ba1.brightsoft.local

Type

A Record

Address

192.168.0.130

Add

Save

Remove

No.	Name	Type	Detail
0	ba1.brightsoft.local	A Record	192.168.0.130
1	dev1.brightsoft.local	A Record	192.168.0.66
2	prod.brightsoft.local	A Record	192.168.0.14
3	staging.brightsoft.local	A Record	192.168.0.241
4	syslog.brightsoft.local	A Record	192.168.0.15

DNS Cache

Top

Question 2

2.1

VLAN Brief

```
SW_F2_L3#SHOW VLAN BRIEF
```

VLAN	Name	Status	Ports
1	default	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
30	Test	active	Fa0/1, Fa0/2
40	UI	active	Fa0/3, Fa0/4
50	VLAN0050	active	Fa0/5, Fa0/6
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW_F2_L3#
```

```
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none
```

```
Name: Fa0/2
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: negotiated
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 30 (Test)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none
```

```
Name: Fa0/3
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: negotiated
```

Name: Fa0/3
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: negotiated
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 40 (UI)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none

Name: Fa0/4
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: negotiated
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 40 (UI)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none

Name: Fa0/5
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: negotiated
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 50 (VLAN0050)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none

•

```
Name: Fa0/6
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: negotiated
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 50 (VLAN0050)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none
```

Interface Switchport

```
SW_F2_L3#SHOW INTERFACES SWITCHPORT
Name: Fa0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: negotiated
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 30 (Test)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
--More--
```

Interfaces Trunk

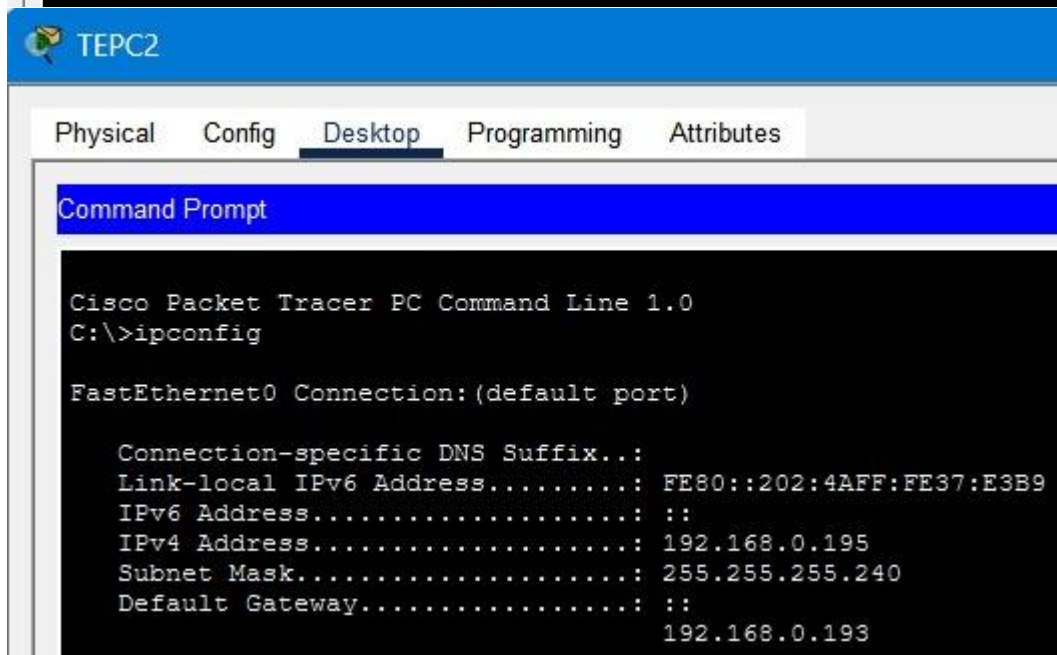
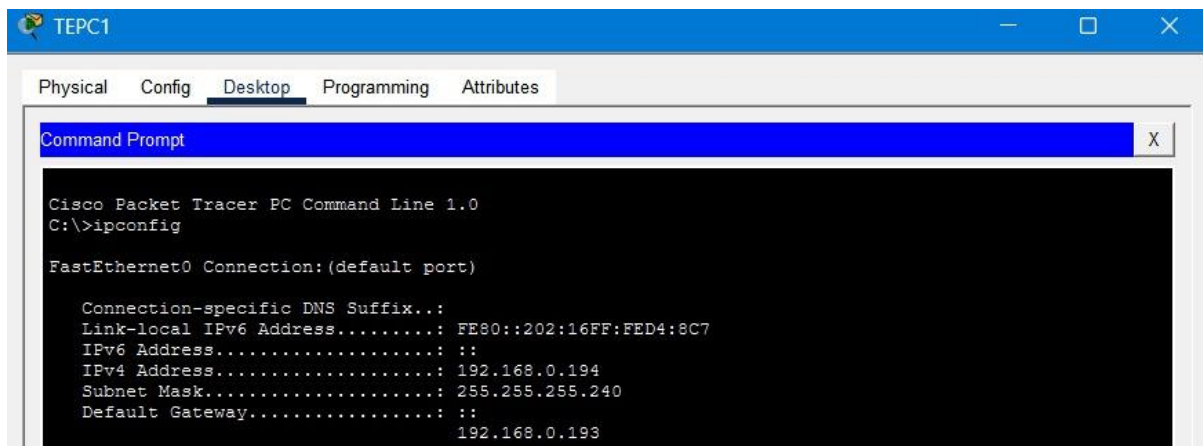
```
SW_F2_L3#SHOW INTERFACES TRUNK
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    1

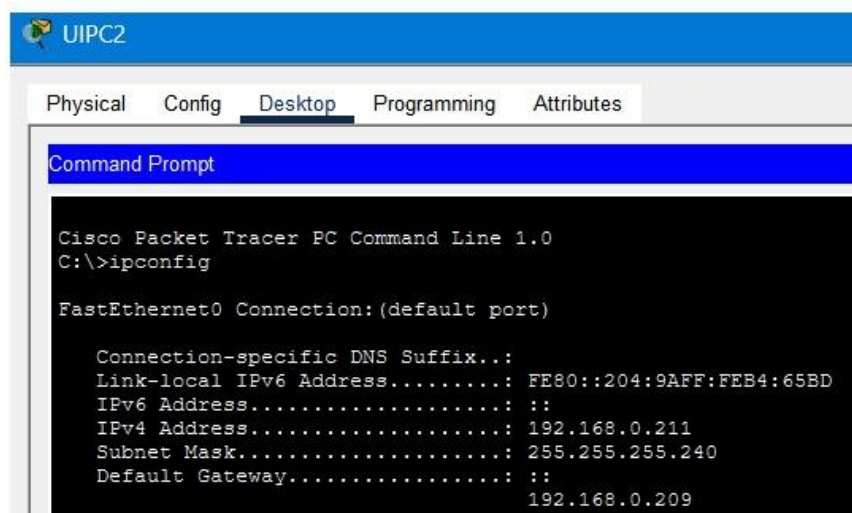
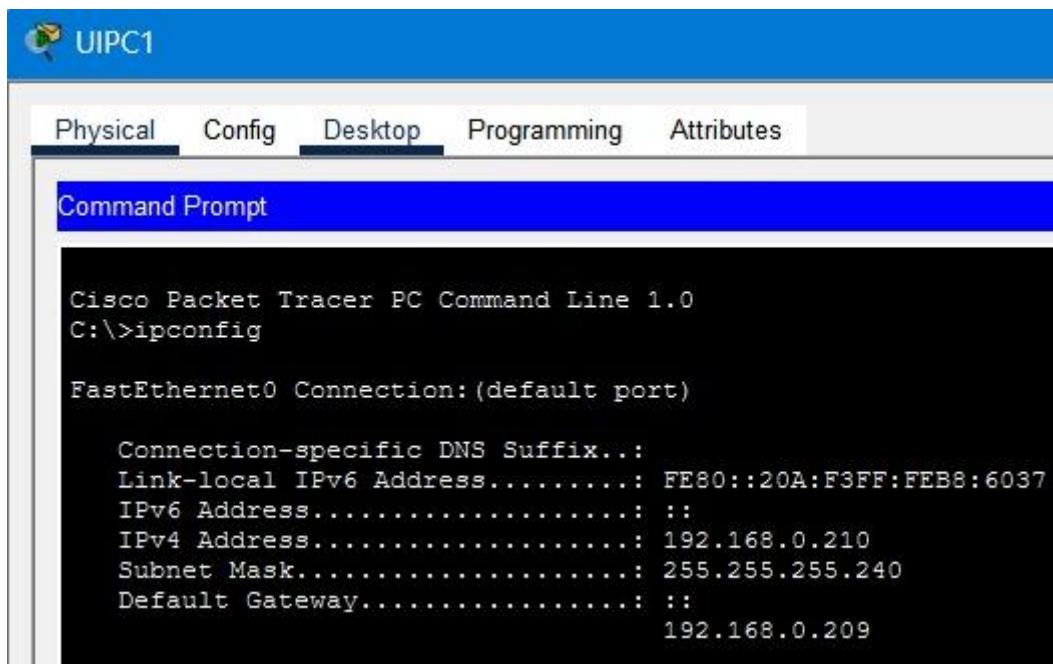
Port      Vlans allowed on trunk
Gig0/1    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,30,40,50

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    1,30,40,50
```

VLAN Verification





Switch IP Interface Brief

```

SW_F2_L3#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
FastEthernet0/1    unassigned      YES unset  up          up
FastEthernet0/2    unassigned      YES unset  up          up
FastEthernet0/3    unassigned      YES unset  up          up
FastEthernet0/4    unassigned      YES unset  up          up
FastEthernet0/5    unassigned      YES unset  up          up
FastEthernet0/6    unassigned      YES unset  up          up
FastEthernet0/7    unassigned      YES unset  down        down
FastEthernet0/8    unassigned      YES unset  down        down
FastEthernet0/9    unassigned      YES unset  down        down
FastEthernet0/10   unassigned      YES unset  down        down
FastEthernet0/11   unassigned      YES unset  down        down
FastEthernet0/12   unassigned      YES unset  down        down
FastEthernet0/13   unassigned      YES unset  down        down
FastEthernet0/14   unassigned      YES unset  down        down
FastEthernet0/15   unassigned      YES unset  down        down
FastEthernet0/16   unassigned      YES unset  down        down
FastEthernet0/17   unassigned      YES unset  down        down
FastEthernet0/18   unassigned      YES unset  down        down
FastEthernet0/19   unassigned      YES unset  down        down
FastEthernet0/20   unassigned      YES unset  down        down
FastEthernet0/21   unassigned      YES unset  down        down
--More--
FastEthernet0/1    unassigned      YES unset  up          up
FastEthernet0/2    unassigned      YES unset  up          up
FastEthernet0/3    unassigned      YES unset  up          up
FastEthernet0/4    unassigned      YES unset  up          up
FastEthernet0/5    unassigned      YES unset  up          up
FastEthernet0/6    unassigned      YES unset  up          up
FastEthernet0/7    unassigned      YES unset  down        down
FastEthernet0/8    unassigned      YES unset  down        down
FastEthernet0/9    unassigned      YES unset  down        down
FastEthernet0/10   unassigned      YES unset  down        down
FastEthernet0/11   unassigned      YES unset  down        down
FastEthernet0/12   unassigned      YES unset  down        down
FastEthernet0/13   unassigned      YES unset  down        down
FastEthernet0/14   unassigned      YES unset  down        down
FastEthernet0/15   unassigned      YES unset  down        down
FastEthernet0/16   unassigned      YES unset  down        down
FastEthernet0/17   unassigned      YES unset  down        down
FastEthernet0/18   unassigned      YES unset  down        down
FastEthernet0/19   unassigned      YES unset  down        down
FastEthernet0/20   unassigned      YES unset  down        down
FastEthernet0/21   unassigned      YES unset  down        down
FastEthernet0/22   unassigned      YES unset  down        down
FastEthernet0/23   unassigned      YES unset  down        down
FastEthernet0/24   unassigned      YES unset  down        down
GigabitEthernet0/1 unassigned      YES unset  up          up
GigabitEthernet0/2 unassigned      YES unset  down        down
Vlan1              unassigned      YES unset  administratively down down
Vlan30             192.168.0.193  YES manual up          up
Vlan40             192.168.0.209  YES manual up          up
Vlan50             192.168.0.225  YES manual up          up

```

Router IP Interface Brief

R_SHARED#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	10.0.0.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	up	up
GigabitEthernet0/1.30	192.168.0.194	YES	manual	up	up
GigabitEthernet0/1.40	192.168.0.210	YES	manual	up	up
GigabitEthernet0/1.50	192.168.0.242	YES	manual	up	up
GigabitEthernet0/2	192.168.1.1	YES	manual	up	up
FastEthernet0/2/0	unassigned	YES	unset	up	up
FastEthernet0/2/1	unassigned	YES	unset	down	down
FastEthernet0/2/2	unassigned	YES	unset	down	down
FastEthernet0/2/3	unassigned	YES	unset	down	down
Serial0/3/0	203.0.113.2	YES	manual	up	down
Serial0/3/1	192.168.2.2	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

Running Configuration

```
!
interface FastEthernet0/1
  switchport access vlan 30
  switchport mode access
!
interface FastEthernet0/2
  switchport access vlan 30
  switchport mode access
!
interface FastEthernet0/3
  switchport access vlan 40
  switchport mode access
!
interface FastEthernet0/4
  switchport access vlan 40
  switchport mode access
!
interface FastEthernet0/5
  switchport access vlan 50
  switchport mode access
!
interface FastEthernet0/6
  switchport access vlan 50
  switchport mode access
!
interface GigabitEthernet0/1
  switchport trunk encapsulation dot1q
  switchport mode trunk
!
interface GigabitEthernet0/2
!
interface Vlan1
  no ip address
  shutdown
!
interface Vlan30
  mac-address 0060.4753.5701
  ip address 192.168.0.193 255.255.255.240
!
interface Vlan40
  mac-address 0060.4753.5702
  ip address 192.168.0.209 255.255.255.240
!
interface Vlan50
  mac-address 0060.4753.5703
  ip address 192.168.0.225 255.255.255.240
!
ip classless
!
ip flow-export version 9
```

IP Route

```
SW_F2_L3#show ip route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
      192.168.0.0/28 is subnetted, 3 subnets
C      192.168.0.192 is directly connected, Vlan30
C      192.168.0.208 is directly connected, Vlan40
C      192.168.0.224 is directly connected, Vlan50
```

Interface Brief

```

SW_F2_L3#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/1 unassigned      YES unset  up          up
FastEthernet0/2 unassigned      YES unset  up          up
FastEthernet0/3 unassigned      YES unset  up          up
FastEthernet0/4 unassigned      YES unset  up          up
FastEthernet0/5 unassigned      YES unset  up          up
FastEthernet0/6 unassigned      YES unset  up          up
FastEthernet0/7 unassigned      YES unset  down        down
FastEthernet0/8 unassigned      YES unset  down        down
FastEthernet0/9 unassigned      YES unset  down        down
FastEthernet0/10 unassigned      YES unset  down        down
FastEthernet0/11 unassigned      YES unset  down        down
FastEthernet0/12 unassigned      YES unset  down        down
FastEthernet0/13 unassigned      YES unset  down        down
FastEthernet0/14 unassigned      YES unset  down        down
FastEthernet0/15 unassigned      YES unset  down        down
FastEthernet0/16 unassigned      YES unset  down        down
FastEthernet0/17 unassigned      YES unset  down        down
FastEthernet0/18 unassigned      YES unset  down        down
FastEthernet0/19 unassigned      YES unset  down        down
FastEthernet0/20 unassigned      YES unset  down        down
FastEthernet0/21 unassigned      YES unset  down        down
..
SW_F2_L3#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/1 unassigned      YES unset  up          up
FastEthernet0/2 unassigned      YES unset  up          up
FastEthernet0/3 unassigned      YES unset  up          up
FastEthernet0/4 unassigned      YES unset  up          up
FastEthernet0/5 unassigned      YES unset  up          up
FastEthernet0/6 unassigned      YES unset  up          up
FastEthernet0/7 unassigned      YES unset  down        down
FastEthernet0/8 unassigned      YES unset  down        down
FastEthernet0/9 unassigned      YES unset  down        down
FastEthernet0/10 unassigned      YES unset  down        down
FastEthernet0/11 unassigned      YES unset  down        down
FastEthernet0/12 unassigned      YES unset  down        down
FastEthernet0/13 unassigned      YES unset  down        down
FastEthernet0/14 unassigned      YES unset  down        down
FastEthernet0/15 unassigned      YES unset  down        down
FastEthernet0/16 unassigned      YES unset  down        down
FastEthernet0/17 unassigned      YES unset  down        down
FastEthernet0/18 unassigned      YES unset  down        down
FastEthernet0/19 unassigned      YES unset  down        down
FastEthernet0/20 unassigned      YES unset  down        down
FastEthernet0/21 unassigned      YES unset  down        down
FastEthernet0/22 unassigned      YES unset  down        down
FastEthernet0/23 unassigned      YES unset  down        down
FastEthernet0/24 unassigned      YES unset  down        down
GigabitEthernet0/1 unassigned      YES unset  up          up
GigabitEthernet0/2 unassigned      YES unset  down        down
Vlan1          unassigned      YES unset  administratively down down
Vlan30         192.168.0.193   YES manual up          up
Vlan40         192.168.0.209   YES manual up          up
Vlan50         192.168.0.225   YES manual up          up

```

2.3

Switch Access Lists

```
SW_F2_L3#show access-lists
Extended IP access list UI_TO_TEST_FILTER
 10 permit ip 192.168.0.160 0.0.0.31 host 192.168.0.134
 20 permit ip 192.168.0.160 0.0.0.31 host 192.168.0.135
 30 deny ip 192.168.0.160 0.0.0.31 192.168.0.128 0.0.0.31
 40 permit ip any any
```

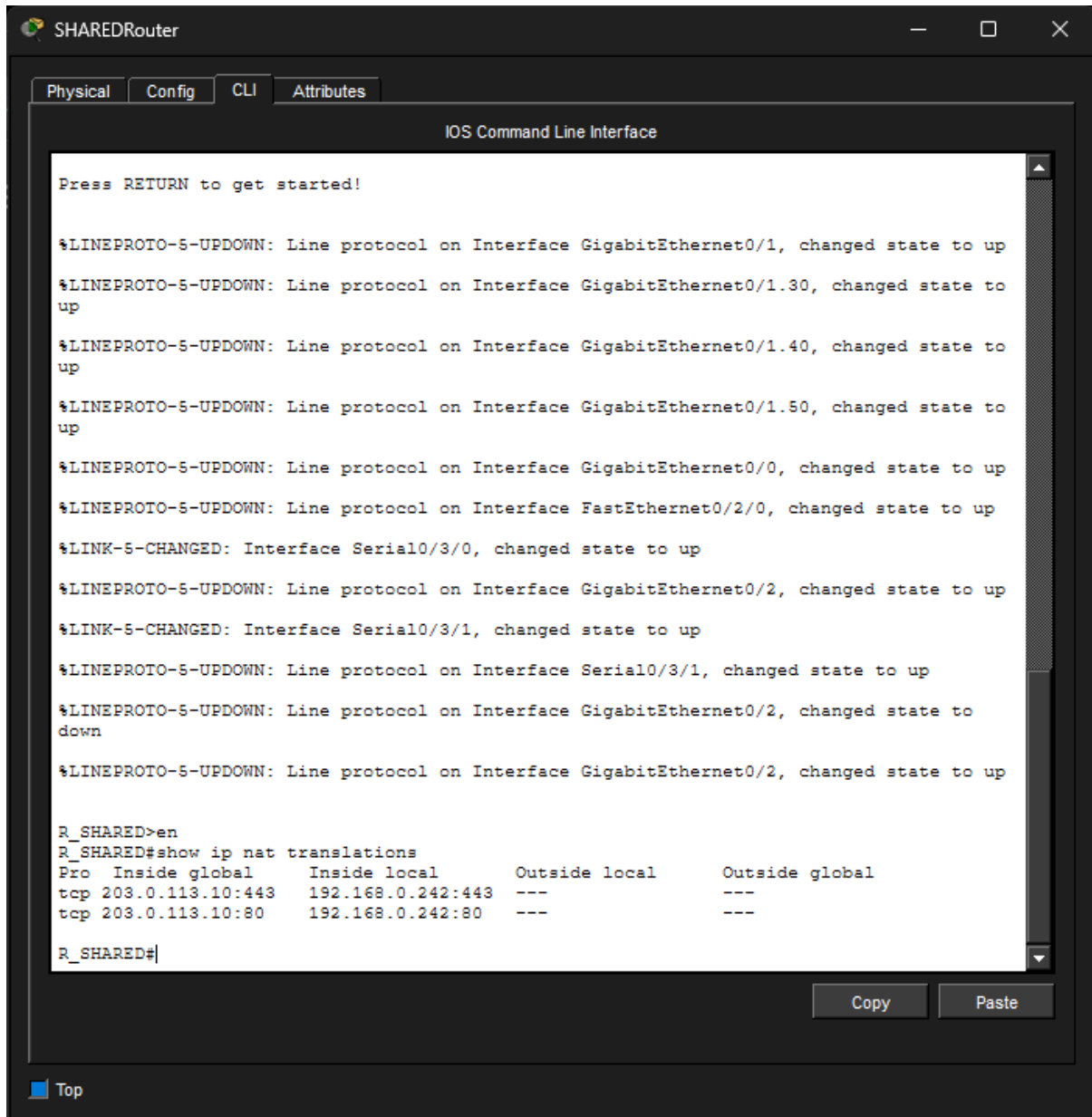
Router Access Lists

```
R_SHARED#show access-lists
Extended IP access list TEST_TO_UI
 10 permit tcp 192.168.0.192 0.0.0.15 host 192.168.0.225
 20 permit tcp 192.168.0.192 0.0.0.15 host 192.168.0.226
 30 permit udp 192.168.0.192 0.0.0.15 host 192.168.0.13 eq domain
 40 permit tcp 192.168.0.192 0.0.0.15 host 192.168.0.241 eq www
 50 permit tcp 192.168.0.192 0.0.0.15 host 192.168.0.241 eq 443
 60 deny ip 192.168.0.192 0.0.0.15 192.168.0.208 0.0.0.15
 70 permit ip any any (4 matches)
Extended IP access list 110
 10 permit tcp host 198.51.100.50 host 203.0.113.10 eq www
 20 permit tcp host 198.51.100.50 host 203.0.113.10 eq 443
 30 deny ip any host 203.0.113.10
 40 permit ip any any
Extended IP access list 111
 10 deny ip 192.168.0.240 0.0.0.15 192.168.0.64 0.0.0.63
 20 deny ip 192.168.0.240 0.0.0.15 192.168.0.128 0.0.0.63
 30 permit ip any any
```

Question 3

3.1

NAT Verification



The screenshot shows the SHAREDRouter application window with the CLI tab selected. The title bar reads "SHAREDRouter". Below the tabs (Physical, Config, CLI, Attributes), the window title is "IOS Command Line Interface". The main text area displays the following output:

```
Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.40, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.50, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2/0, changed state to up
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

R_SHARED>en
R_SHARED#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
tcp 203.0.113.10:443    192.168.0.242:443 ---                ---
tcp 203.0.113.10:80     192.168.0.242:80  ---                ---

R_SHARED#
```

At the bottom right of the CLI window are "Copy" and "Paste" buttons. At the bottom left of the application window is a "Top" button.

SHAREDRouter

Physical

Config

CLI

Attributes

IOS Command Line Interface

```
up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.50, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

R_SHARED>en
R_SHARED#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
tcp  203.0.113.10:443    192.168.0.242:443 ---                ---
tcp  203.0.113.10:80     192.168.0.242:80  ---                ---

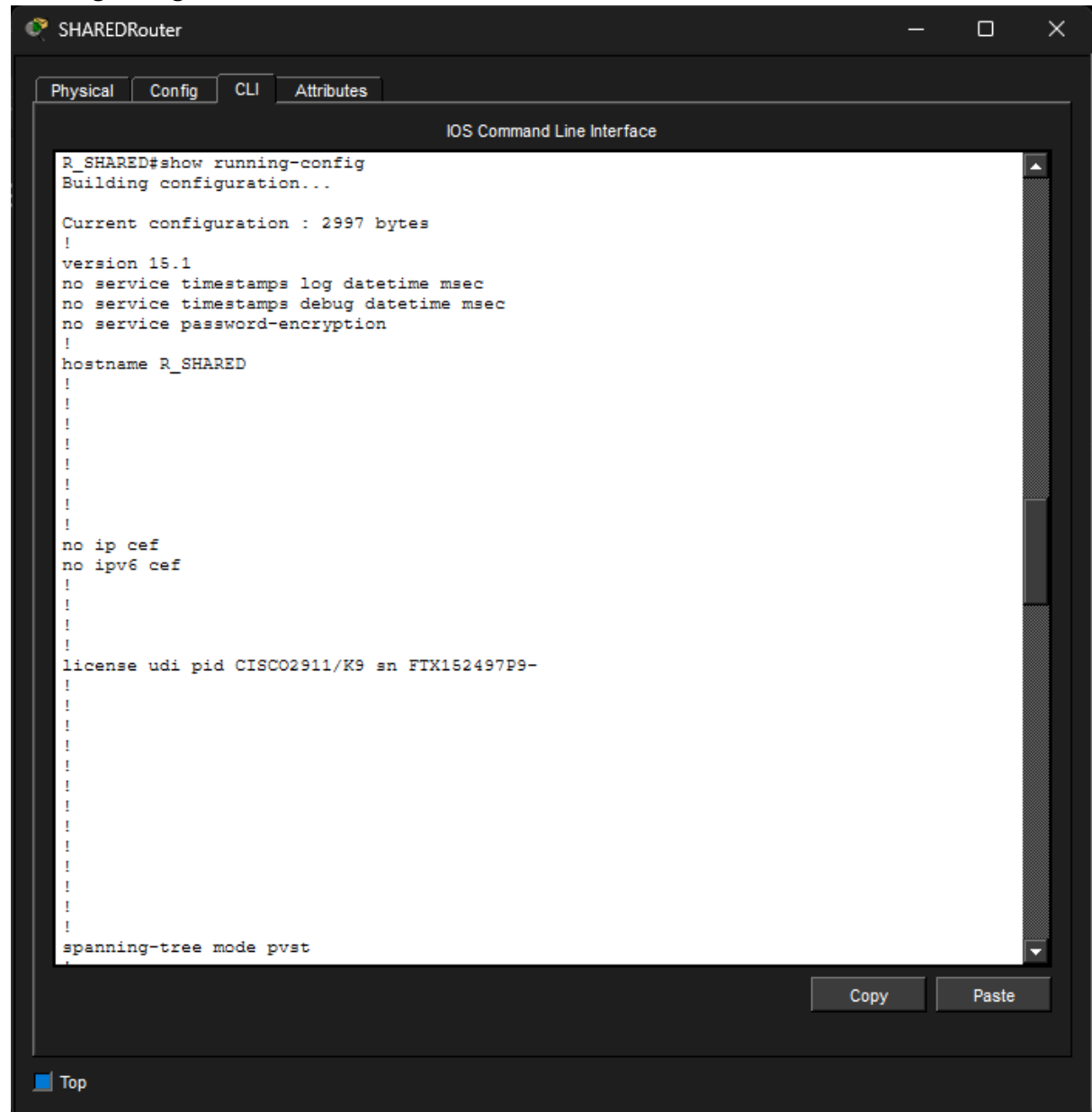
R_SHARED#show ip nat statistics
Total translations: 2 (4 static, 4294967294 dynamic, 2 extended)
Outside Interfaces: Serial0/3/0
Inside Interfaces: GigabitEthernet0/0 , GigabitEthernet0/1.30 , GigabitEthernet0/1.50
Hits: 0 Misses: 0
Expired translations: 0
Dynamic mappings:
R_SHARED#
```

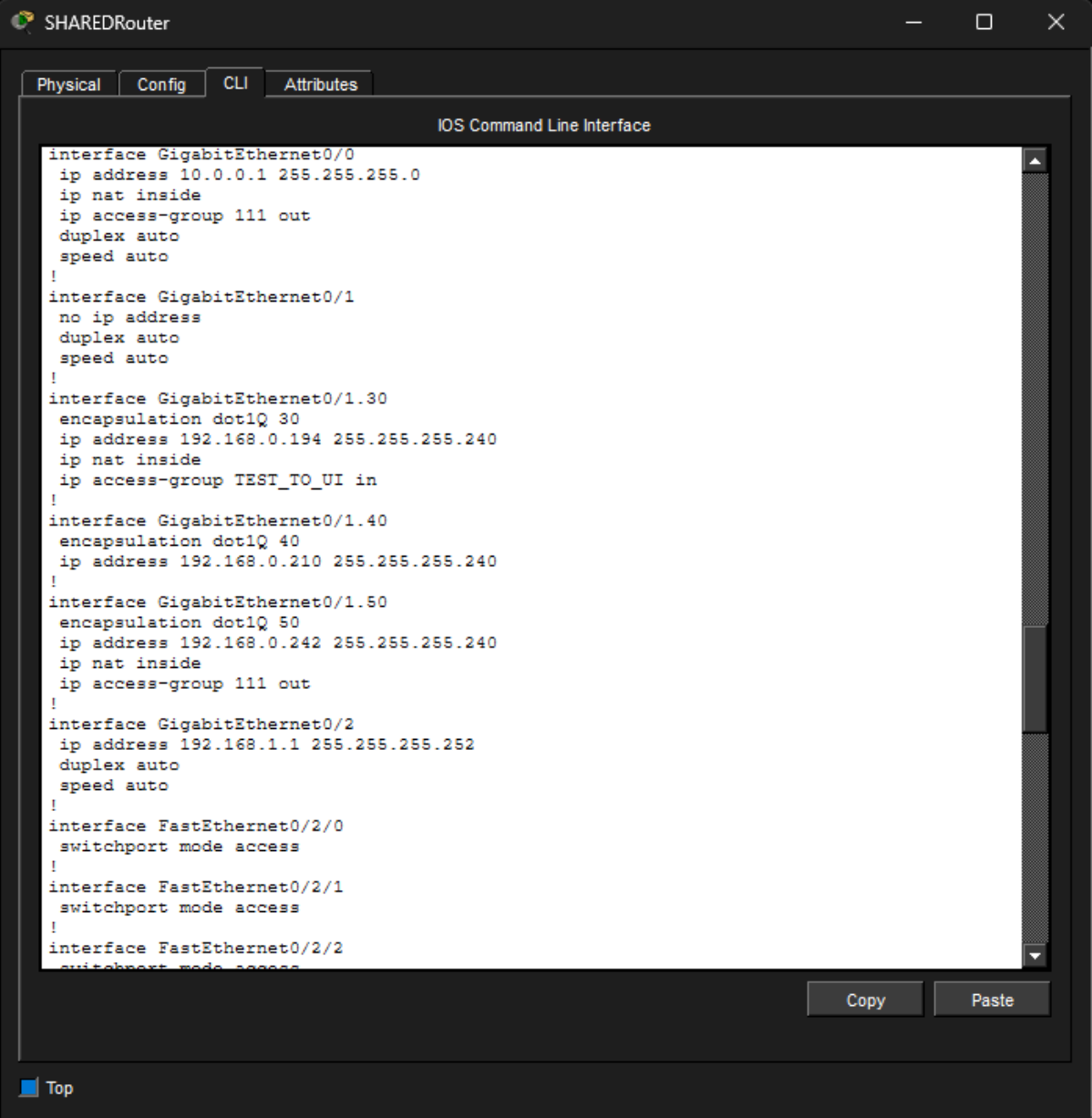
Copy

Paste

Top

Running Configuration





IOS Command Line Interface

```
interface FastEthernet0/2/2
 switchport mode access
!
interface FastEthernet0/2/3
 switchport mode access
!
interface Serial0/3/0
 ip address 203.0.113.2 255.255.255.252
 ip nat outside
 ip access-group 110 in
 clock rate 2000000
!
interface Serial0/3/1
 ip address 192.168.2.2 255.255.255.252
 clock rate 2000000
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 log-adjacency-changes
 redistribute rip metric 100 subnets
 network 192.168.0.192 0.0.0.15 area 0
 network 192.168.0.208 0.0.0.15 area 0
 network 192.168.0.224 0.0.0.15 area 0
 network 192.168.0.240 0.0.0.15 area 0
 network 192.168.2.0 0.0.0.3 area 0
!
router rip
 version 2
 network 192.168.0.0
 no auto-summary
!
ip nat inside source static tcp 192.168.0.241 80 203.0.113.10 80
ip nat inside source static tcp 192.168.0.241 443 203.0.113.10 443
ip nat inside source static tcp 192.168.0.242 80 203.0.113.10 80
ip nat inside source static tcp 192.168.0.242 443 203.0.113.10 443
ip classless
ip route 0.0.0.0 0.0.0.0 Serial0/3/0
```

Copy

Paste

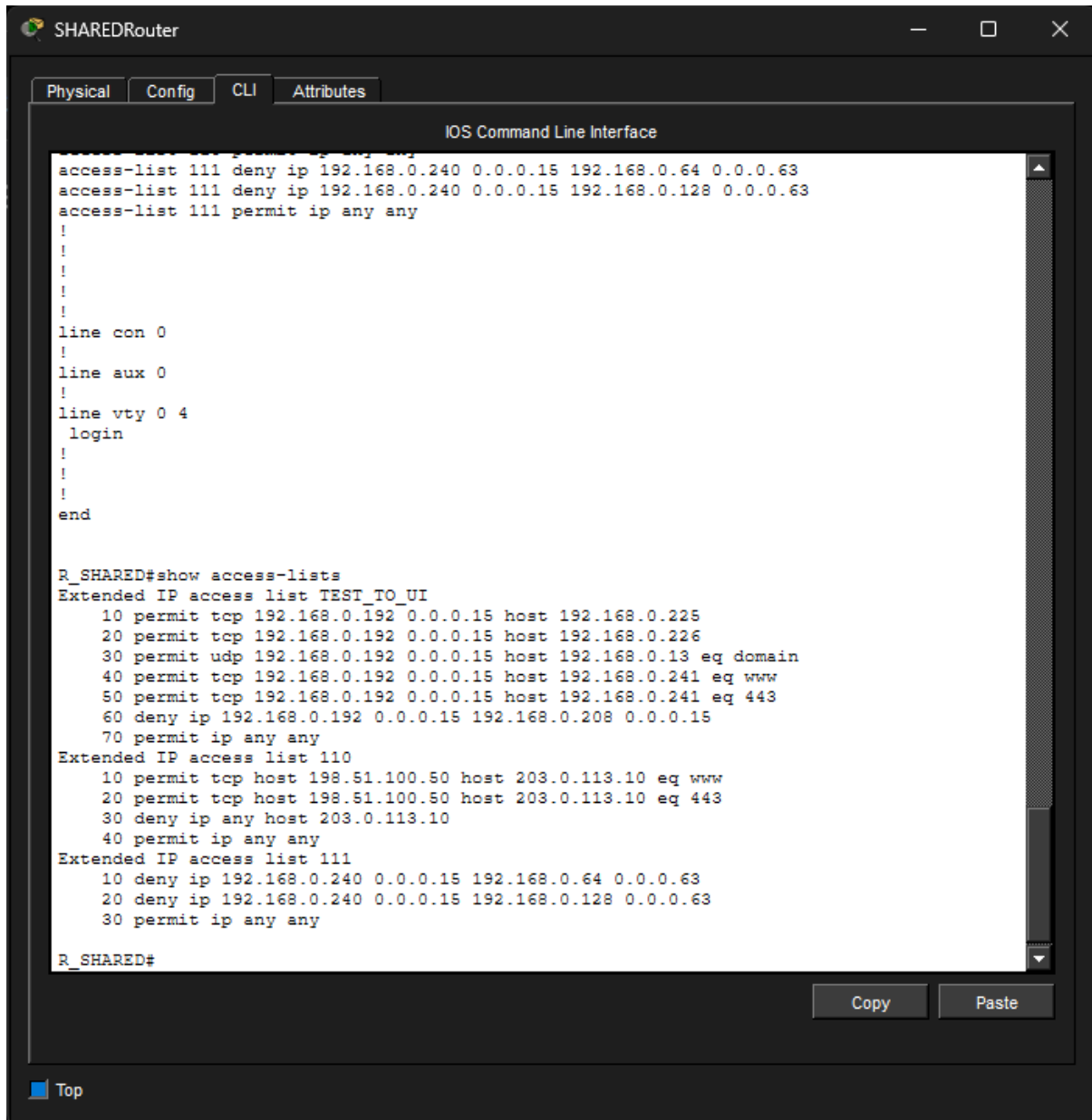
IOS Command Line Interface

```
ip nat inside source static tcp 192.168.0.241 80 203.0.113.10 80
ip nat inside source static tcp 192.168.0.241 443 203.0.113.10 443
ip nat inside source static tcp 192.168.0.242 80 203.0.113.10 80
ip nat inside source static tcp 192.168.0.242 443 203.0.113.10 443
ip classless
ip route 0.0.0.0 0.0.0.0 Serial0/3/0
!
ip flow-export version 9
!
!
ip access-list extended TEST_TO_UI
  permit tcp 192.168.0.192 0.0.0.15 host 192.168.0.225
  permit tcp 192.168.0.192 0.0.0.15 host 192.168.0.226
  permit udp 192.168.0.192 0.0.0.15 host 192.168.0.13 eq domain
  permit tcp 192.168.0.192 0.0.0.15 host 192.168.0.241 eq www
  permit tcp 192.168.0.192 0.0.0.15 host 192.168.0.241 eq 443
  deny ip 192.168.0.192 0.0.0.15 192.168.0.208 0.0.0.15
  permit ip any any
access-list 110 permit tcp host 198.51.100.50 host 203.0.113.10 eq www
access-list 110 permit tcp host 198.51.100.50 host 203.0.113.10 eq 443
access-list 110 deny ip any host 203.0.113.10
access-list 110 permit ip any any
access-list 111 deny ip 192.168.0.240 0.0.0.15 192.168.0.64 0.0.0.63
access-list 111 deny ip 192.168.0.240 0.0.0.15 192.168.0.128 0.0.0.63
access-list 111 permit ip any any
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
  login
!
!
!
end
```

Copy

Paste

Access Lists



The screenshot shows the SHAREDRouter CLI interface with the 'CLI' tab selected. The 'IOS Command Line Interface' window displays the following configuration and verification output:

```
access-list 111 deny ip 192.168.0.240 0.0.0.15 192.168.0.64 0.0.0.63
access-list 111 deny ip 192.168.0.240 0.0.0.15 192.168.0.128 0.0.0.63
access-list 111 permit ip any any
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end

R_SHARED#show access-lists
Extended IP access list TEST_TO_UI
 10 permit tcp 192.168.0.192 0.0.0.15 host 192.168.0.225
 20 permit tcp 192.168.0.192 0.0.0.15 host 192.168.0.226
 30 permit udp 192.168.0.192 0.0.0.15 host 192.168.0.13 eq domain
 40 permit tcp 192.168.0.192 0.0.0.15 host 192.168.0.241 eq www
 50 permit tcp 192.168.0.192 0.0.0.15 host 192.168.0.241 eq 443
 60 deny ip 192.168.0.192 0.0.0.15 192.168.0.208 0.0.0.15
 70 permit ip any any
Extended IP access list 110
 10 permit tcp host 198.51.100.50 host 203.0.113.10 eq www
 20 permit tcp host 198.51.100.50 host 203.0.113.10 eq 443
 30 deny ip any host 203.0.113.10
 40 permit ip any any
Extended IP access list 111
 10 deny ip 192.168.0.240 0.0.0.15 192.168.0.64 0.0.0.63
 20 deny ip 192.168.0.240 0.0.0.15 192.168.0.128 0.0.0.63
 30 permit ip any any

R_SHARED#
```

At the bottom of the CLI window, there are 'Copy' and 'Paste' buttons. A 'Top' button is located at the bottom left of the window frame.

Logging

IOS Command Line Interface

```
R_SHARED#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
                  0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

  Console logging: level debugging, 12 messages logged, xml disabled,
                    filtering disabled
  Monitor logging: level debugging, 12 messages logged, xml disabled,
                    filtering disabled
  Buffer logging:   level debugging, 0 messages logged, xml disabled,
                    filtering disabled

Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

ESM: 0 messages dropped
    Trap logging: level informational, 12 message lines logged

Log Buffer (4096 bytes):

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.30, changed state to
up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.30, changed state to
up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.40, changed state to
```

Copy

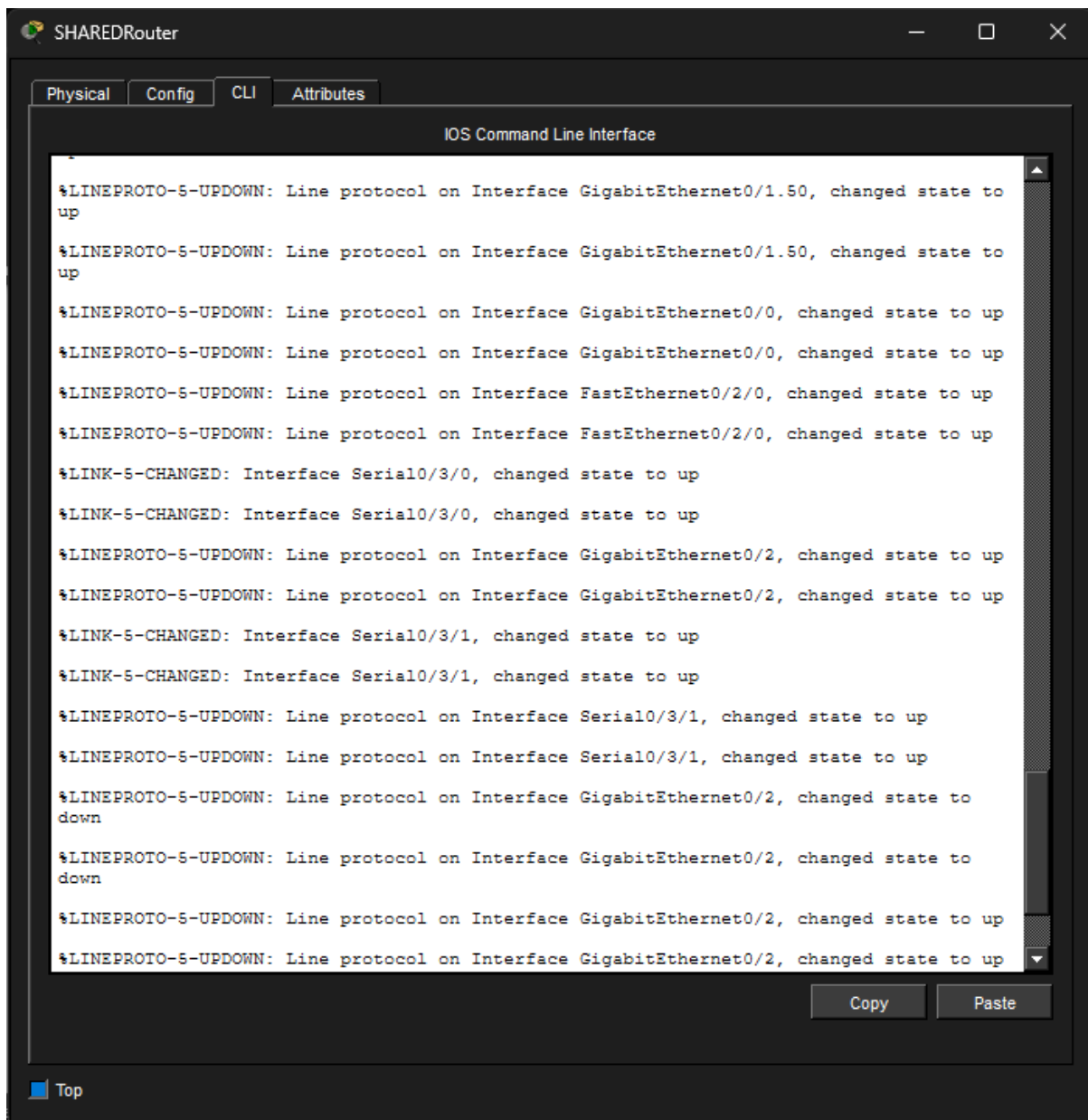
Paste

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.40, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.40, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.50, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.50, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2/0, changed state to up
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to
```

Copy

Paste



3.2

The main challenge of ensuring that a web server is accessible to external patterns while keeping strict internal segmentation and security. Placing a web server on an internal network segment poses a significant risk so it is recommended that the best design is to implement a dedicated demilitarized zone (DMZ)

Placing the staging web server in a dedicated DMZ is the most secure and best approach for the server.

In terms of containment , the DMZ is a separated network that sits between the trusted internal network and the and the untrusted external attackers , thus the DMZ ensures strict access

controls that will stop attackers from lateral movement to the main development , analyst and production networks.

The DMZ ensures that there are clear and set security boundaries set which allows for the implementation of precise firewalls due to the network being trusted, the internet being an untrusted because of it being external.

The DMZ ensures that it meets the operational needs of the organization as it supports the server being accessible from both the external internet and specific internal groups whilst it being restricted from other internal teams.

required Control Plane and Data Policies

this essentially defines the rules of the road for network traffic.

It is essential to have stateful firewall especially a next generation firewalls that should be placed at the network perimeter that ensures it controls all traffic to and from the DMZ and internal network.

Access controls Lists.

In conjunction with the main firewall, interface ACL'S can be applied on the routers and layer 3 switch which will allow for defence in-depth. If there are shared router that are connected to the DMZ this will mean that an inbound ACL on the DMZ interface will block all traffic except the specific ports allowed from the internet and developer subnet.

If it on a specific router that is specific to a group such as "Developers group". there will be an outbound ACL on the interface that faces the DMZ that ensures that only developer's subnet can reach the DMZ. This helps reinforce the principle of least privilege.

Routing Considerations

The DMZ should be advertised into the internal routing protocols by implementing OSPF on floor 2 and via redistribution into RIP for floor 1 which will help ensure internal devices know how to reach it.

In terms of external routing the external partners will need to reach the DMZ as it does not need to be advertised to the internet. The external partners will reach the DMZ via its public IP address through the firewall NAT rules in which the firewall will be able to perform destination NAT that will enable the translation of the public IP to the staging server private DMZ.

In terms of the routing symmetry and design there must be a design with a backup link. Taking that into consideration we must look at firewalls in terms of symmetry as they are symmetrical, meaning that firewalls are stateful and the return traffic for a connection must pass through the same firewall the request was sent through. This will ensure that routing is configured in symmetrical and functional way that allows them to avoid asymmetric routing issues during failover.

Supporting Devices

DNS resolution is the process of converting a human – readable domain name into machine readable IP addresses, this process is essential for enabling communication between devices on the internet or within private networks. This is an approach that should be used for supporting devices. External DNS such as public DNS record should point to the public IP address of the firewall that fronts the DMZ. Internal DNS such as the central DNS server should be configured with a split-horizon DNS zone.

Certificate management which will ensure that the staging server uses valid and publicly trusted TLS/SSL certificates for HTTPS, this will help ensure that security warnings for external partners are prevented during demonstrations. To ensure that

these certificates are constantly up to date by automating the certificate renewal process to prevent them expiring.

Monitoring and logging and Intrusion detection

the staging server and the perimeter firewall must be configured to allow for all logs to be sent to the central Syslog server which includes the web server access and error logs , the firewalls log

of allow/deny especially all connections from the internet and the system authentication logs from the server.

In terms of monitoring, a monitoring system should be able to monitor the staging server's core hardware this being the CPU, memory, disk ext. And should also be able to send alerts on downtime.

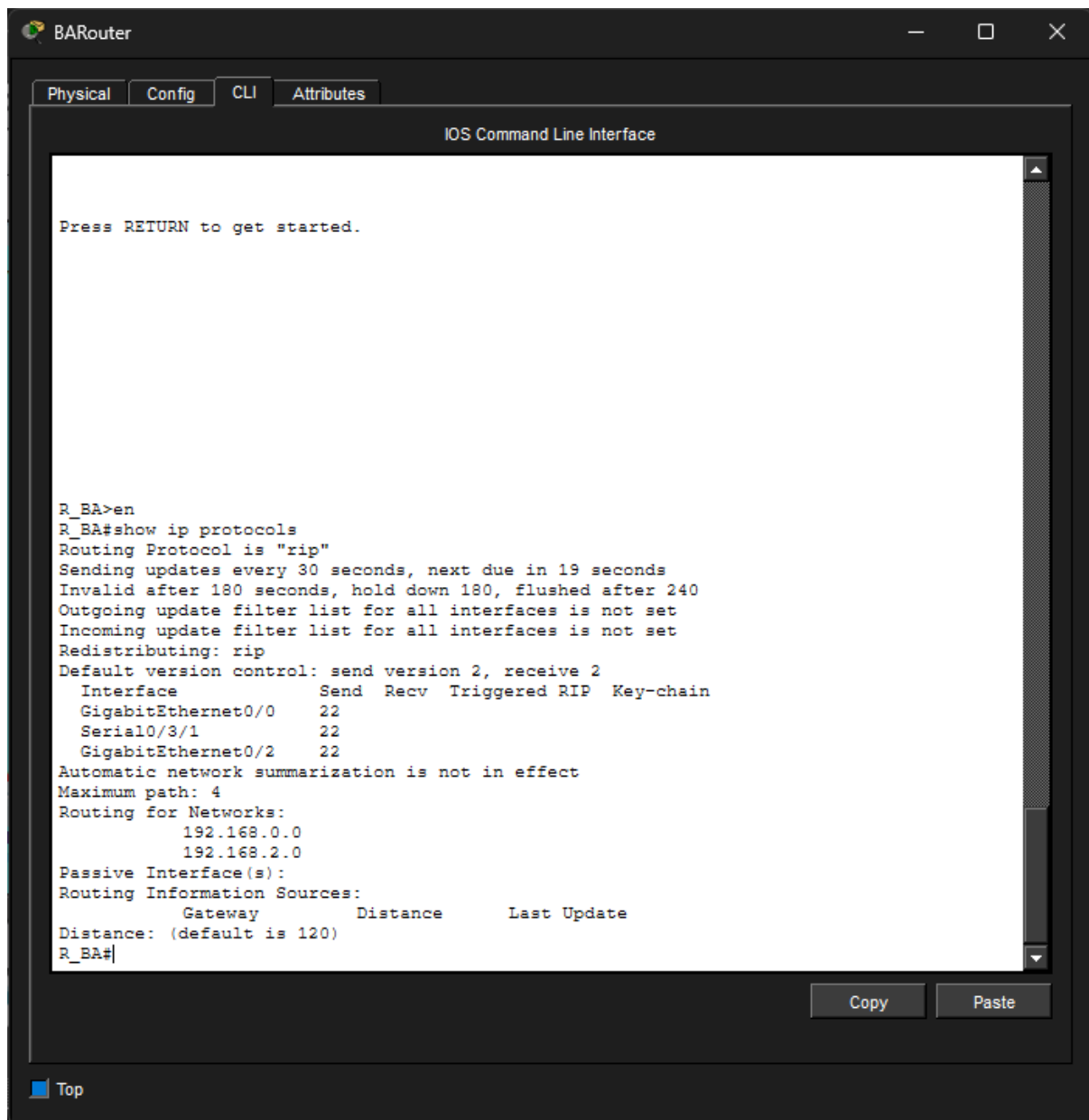
In terms of intrusion detection, the perimeters should be configured with IDS/IPS signature and should constantly scan incoming HTTP/HTTPS traffic from common web application attacks such as SQL injection. IDS mode should be used to avoid blocking updates that come from the developers and move to IPS mode during stable demo periods.

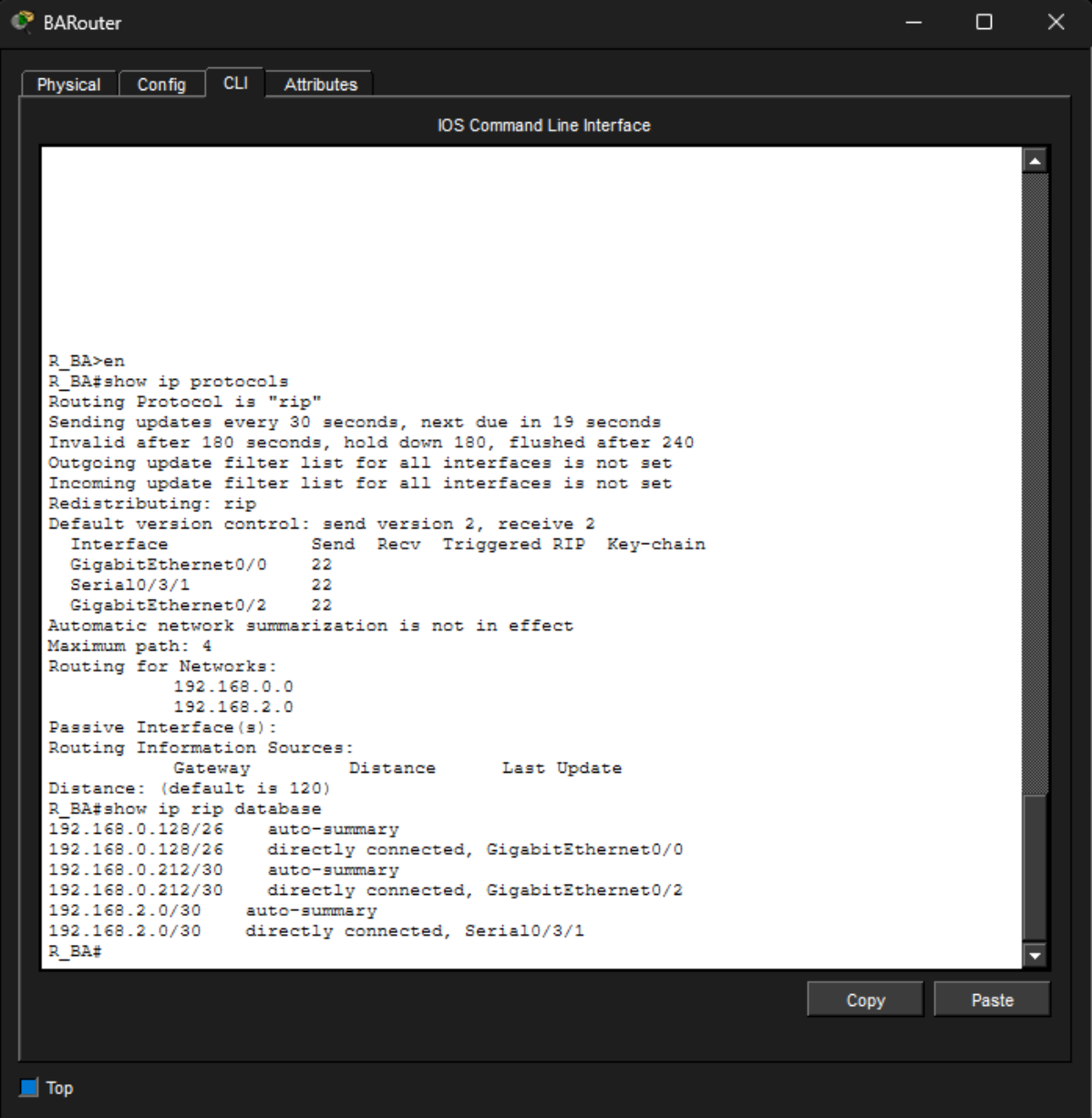
(Fortinet, 2025) (GeeksforGeeks, 2025) (chiradeep-basumallick, 2024) (PaloAlto, 2025)

Question 4

4.1

RIP Verification





IOS Command Line Interface

```
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
    192.168.0.0
    192.168.2.0
Passive Interface(s):
Routing Information Sources:
    Gateway         Distance      Last Update
Distance: (default is 120)
R_BA#show ip rip database
192.168.0.128/26    auto-summary
192.168.0.128/26    directly connected, GigabitEthernet0/0
192.168.0.212/30    auto-summary
192.168.0.212/30    directly connected, GigabitEthernet0/2
192.168.2.0/30      auto-summary
192.168.2.0/30      directly connected, Serial0/3/1
R_BA#show ip route rip

R_BA#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.0.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.0.128/26 is directly connected, GigabitEthernet0/0
L       192.168.0.129/32 is directly connected, GigabitEthernet0/0
C       192.168.0.212/30 is directly connected, GigabitEthernet0/2
L       192.168.0.213/32 is directly connected, GigabitEthernet0/2
    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/30 is directly connected, Serial0/3/1
L       192.168.2.1/32 is directly connected, Serial0/3/1

R_BA#
```

Copy

Paste

OSPF Verification

The screenshot shows the SHAREDRouter application window with the CLI tab selected. The title bar includes a logo, the text "SHAREDRouter", and standard window controls. Below the title bar are tabs for "Physical", "Config", "CLI", and "Attributes". The main area is titled "IOS Command Line Interface" and contains a text box with the following text:

```
R_SHARED#show ip ospf neighbor

R_SHARED#show ip ospf interface

GigabitEthernet0/1.30 is up, line protocol is up
Internet address is 192.168.0.194/28, Area 0
Process ID 1, Router ID 203.0.113.2, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 203.0.113.2, Interface address 192.168.0.194
No backup designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Hello due in 00:00:07
Index 1/1, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 0, Adjacent neighbor count is 0
Suppress hello for 0 neighbor(s)

GigabitEthernet0/1.40 is up, line protocol is up
Internet address is 192.168.0.210/28, Area 0
Process ID 1, Router ID 203.0.113.2, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 203.0.113.2, Interface address 192.168.0.210
No backup designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Hello due in 00:00:06
Index 2/2, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 0, Adjacent neighbor count is 0
Suppress hello for 0 neighbor(s)

GigabitEthernet0/1.50 is up, line protocol is up
Internet address is 192.168.0.242/28, Area 0
Process ID 1, Router ID 203.0.113.2, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 203.0.113.2, Interface address 192.168.0.242
No backup designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Hello due in 00:00:07
Index 2/2, flood queue length 0
```

At the bottom right of the text box are "Copy" and "Paste" buttons. At the bottom left of the window is a "Top" button.

IOS Command Line Interface

```
Suppress hello for 0 neighbor(s)
GigabitEthernet0/1.40 is up, line protocol is up
Internet address is 192.168.0.210/28, Area 0
Process ID 1, Router ID 203.0.113.2, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 203.0.113.2, Interface address 192.168.0.210
No backup designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:06
Index 2/2, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 0, Adjacent neighbor count is 0
Suppress hello for 0 neighbor(s)
GigabitEthernet0/1.50 is up, line protocol is up
Internet address is 192.168.0.242/28, Area 0
Process ID 1, Router ID 203.0.113.2, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 203.0.113.2, Interface address 192.168.0.242
No backup designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:07
Index 3/3, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 0, Adjacent neighbor count is 0
Suppress hello for 0 neighbor(s)
Serial0/3/1 is up, line protocol is up
Internet address is 192.168.2.2/30, Area 0
Process ID 1, Router ID 203.0.113.2, Network Type POINT-TO-POINT, Cost: 64
Transmit Delay is 1 sec, State POINT-TO-POINT,
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:02
Index 4/4, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Suppress hello for 0 neighbor(s)
```

Copy

Paste

SHAREDRouter

Physical
Config
CLI
Attributes

IOS Command Line Interface

```

Index 2/2, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 0, Adjacent neighbor count is 0
Suppress hello for 0 neighbor(s)
GigabitEthernet0/1.50 is up, line protocol is up
Internet address is 192.168.0.242/28, Area 0
Process ID 1, Router ID 203.0.113.2, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 203.0.113.2, Interface address 192.168.0.242
No backup designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Hello due in 00:00:07
Index 3/3, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 0, Adjacent neighbor count is 0
Suppress hello for 0 neighbor(s)
Serial0/3/1 is up, line protocol is up
Internet address is 192.168.2.2/30, Area 0
Process ID 1, Router ID 203.0.113.2, Network Type POINT-TO-POINT, Cost: 64
Transmit Delay is 1 sec, State POINT-TO-POINT,
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Hello due in 00:00:02
Index 4/4, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Suppress hello for 0 neighbor(s)
R_SHARED#
R_SHARED#show ip ospf database
      OSPF Router with ID (203.0.113.2) (Process ID 1)

      Router Link States (Area 0)

Link ID      ADV Router   Age         Seq#         Checksum Link count
203.0.113.2  203.0.113.2  102        0x80000007  0x005e03  4
R_SHARED#

```

CopyPaste

Top

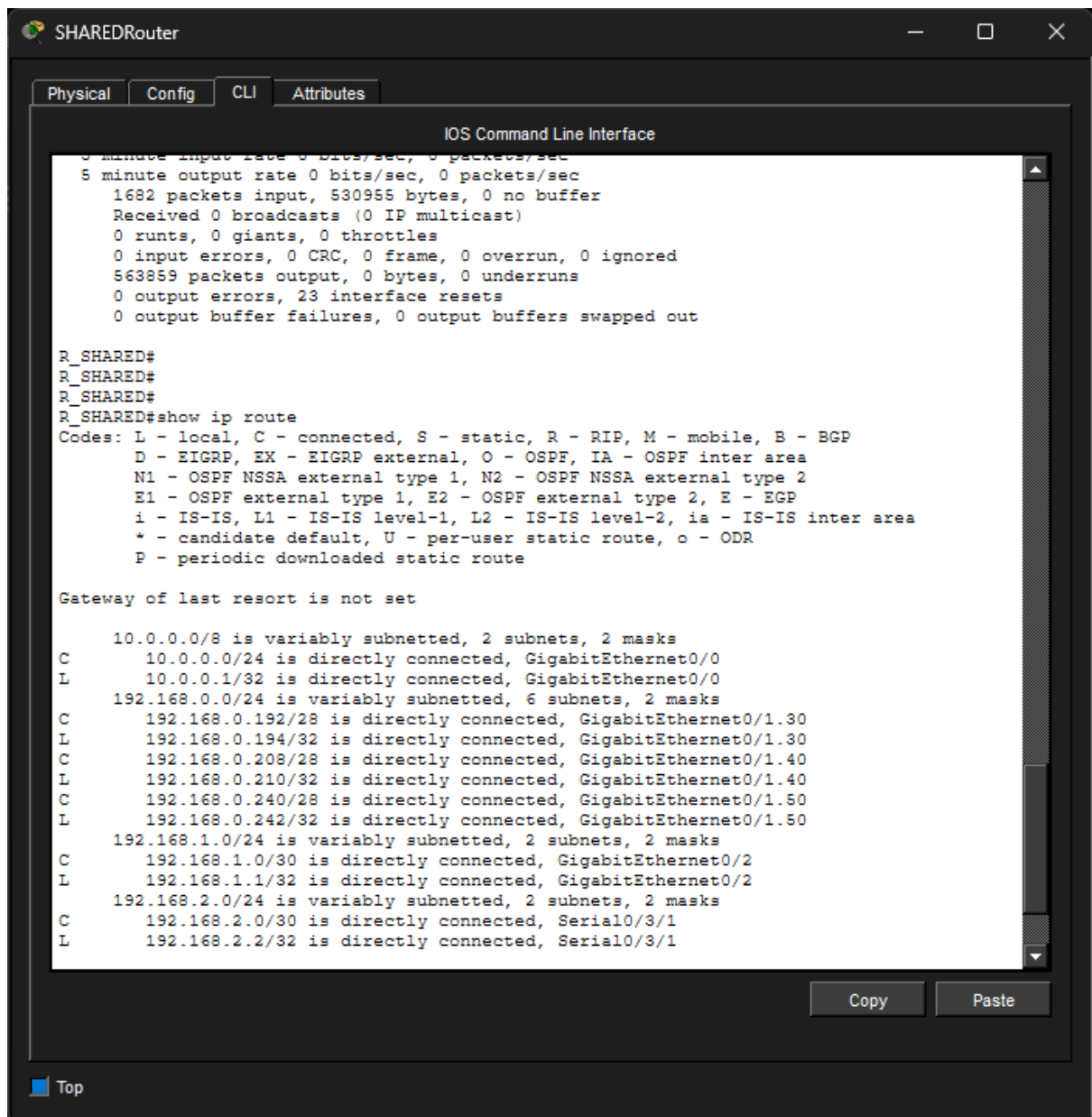
Backup link

Shared Router

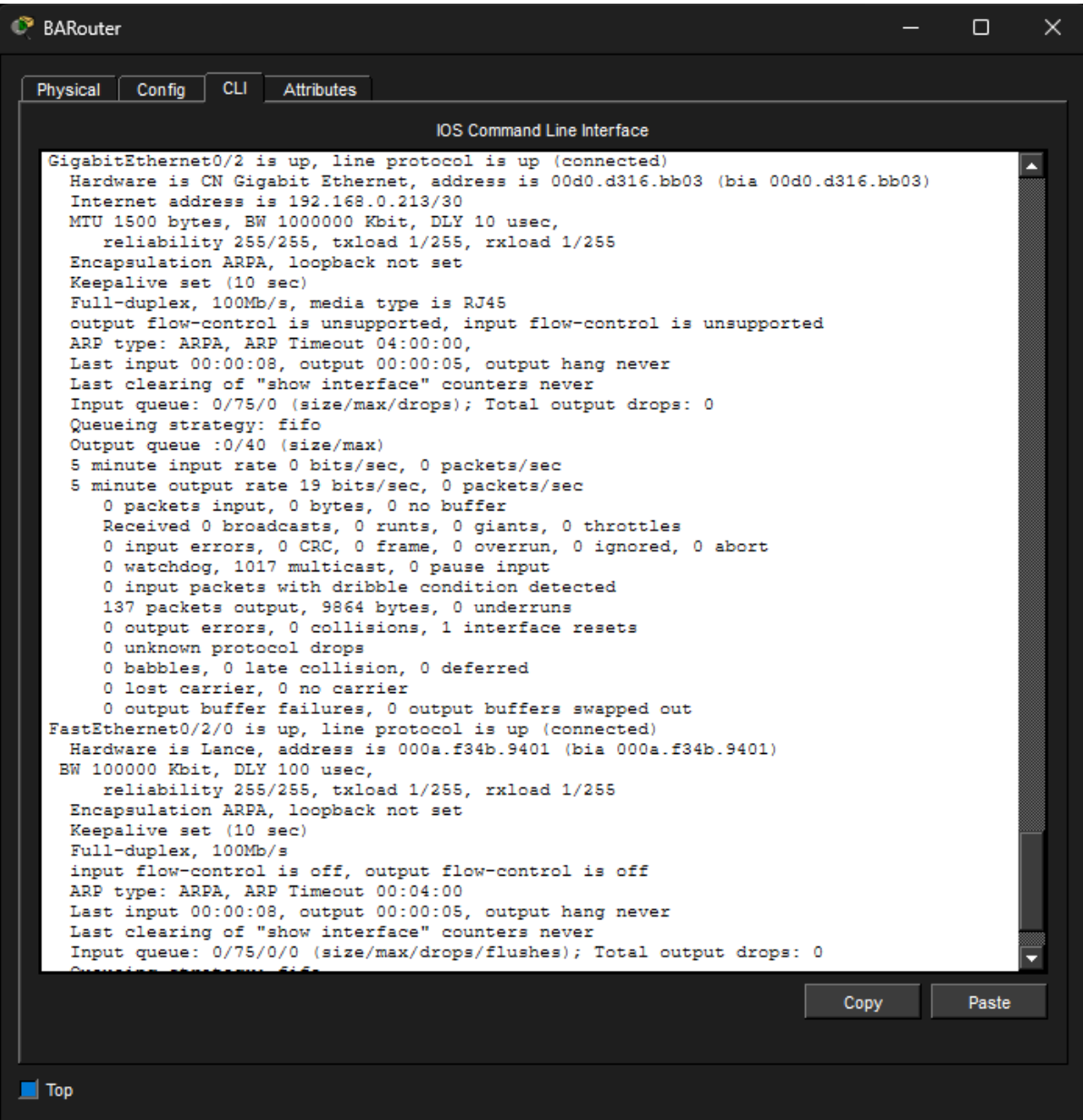
The screenshot shows the SHAREDRouter application window. The 'CLI' tab is selected, displaying the 'IOS Command Line Interface'. The output shows the status of two network interfaces:

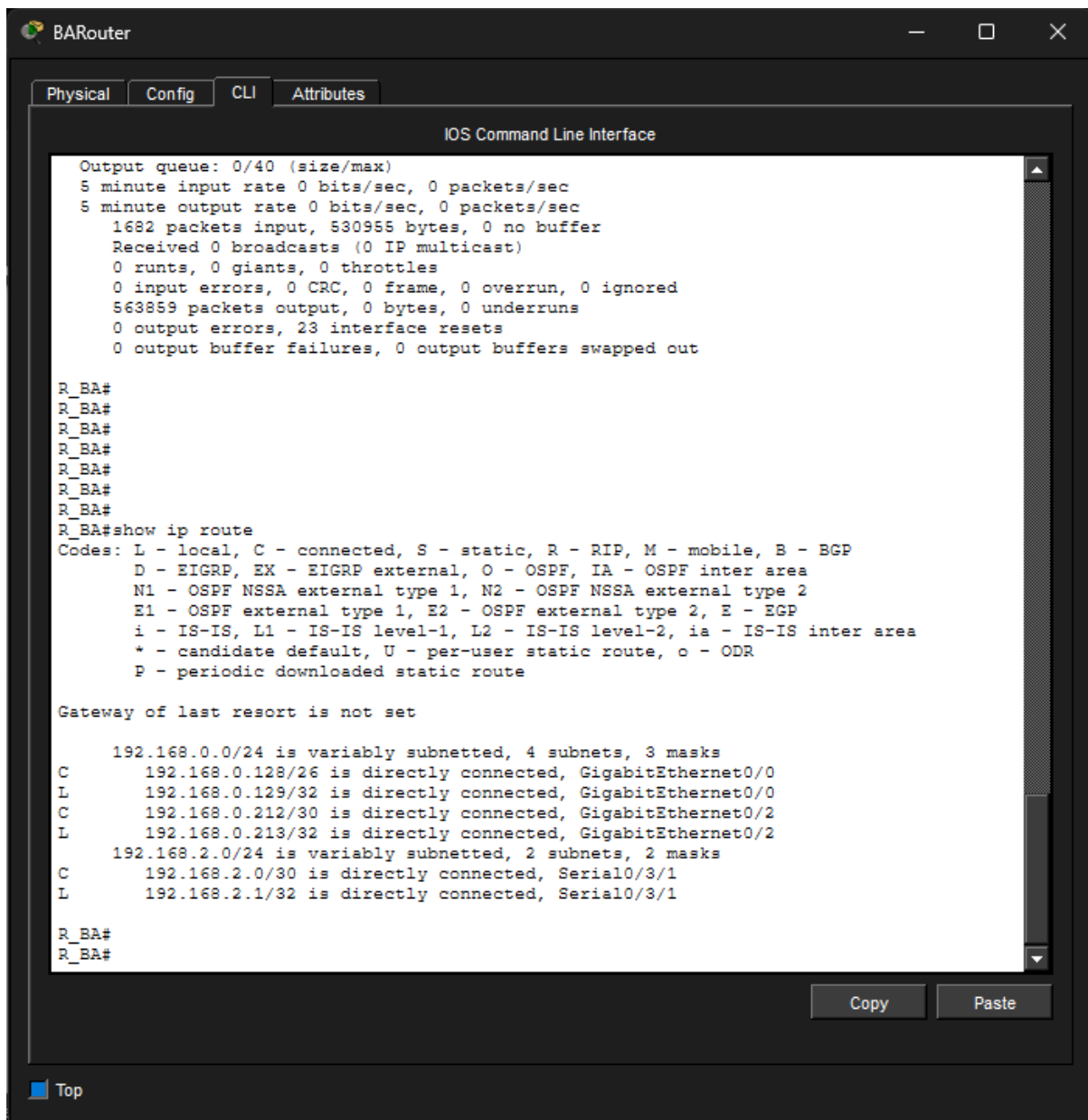
```
GigabitEthernet0/2 is up, line protocol is up (connected)
Hardware is CN Gigabit Ethernet, address is 0010.1122.0103 (bia 0010.1122.0103)
Internet address is 192.168.1.1/30
MTU 1500 bytes, BW 1000000 Kbit, DLY 10 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s, media type is RJ45
output flow-control is unsupported, input flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00,
Last input 00:00:08, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0 (size/max/drops); Total output drops: 0
Queueing strategy: fifo
Output queue :0/40 (size/max)
5 minute input rate 21 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
    145 packets input, 10380 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 1017 multicast, 0 pause input
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
FastEthernet0/2/0 is up, line protocol is up (connected)
Hardware is Lance, address is 0001.c7ad.ab01 (bia 0001.c7ad.ab01)
BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s
input flow-control is off, output flow-control is off
ARP type: ARPA, ARP Timeout 00:04:00
Last input 00:00:08, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
```

At the bottom of the CLI window, there are 'Copy' and 'Paste' buttons. Below the CLI window, there is a 'Top' button.



BA Router





4.2

Challenges of Running RIP and OSPF in a hybrid Environment

Running RIP and OSPF creates a hybrid environment, but it also poses its challenges, this is due to this design introducing several challenges related to protocol fundamentals, management and network resilience.

Protocol Compatibility

The challenges from having RIP and OSPF comes from the basis of how fundamentally different they are from each other.

In terms of metric Incompatibility:

RIP (Routing Information Protocol) uses a hop count as its only metric. a hop count of 16 is unreachable and gives an extremely limited view of the network.

OSPF (Open Shortest Path First) this uses Cost, a value that is derived from bandwidth e.g. 100 Mbps link 1 Cost. This allows for a more intelligent and efficient path choice.

The problem with having these two protocols is that there is no direct conversion between hop count and cost meaning that a network administrator that faces this challenge must manually assign a seed metric during translation.

In terms of Convergence time:

RIP has a slow convergence protocol as it uses timers which cause it to have a slower recovery from network failures, thus increasing the risk of temporary routing loops. OSPF has a fast convergence protocol as it uses a complex state mechanism, and it triggers updates at once upon a topology change.

The challenge of convergence Time is that during network events the OSPF domain will have reconverged on a new path long before the RIP domain has stabilised. This will cause intermittent connectivity and will leave an unpredictable routing behaviour between the floors.

In terms of network size and scalability

The OSPF is highly scalable and supports large hierarchical enterprise networks through the use of areas. The RIP has a max hop count of 15 which limits the diameter of a network. This will hinder future growth.

The challenges with both the OSPF and RIP in terms of network size and scalability are that the whole network is limited by the weakest link, this being the RIP protocol's limited hop count of 15. The nature of the RIP means that it cannot summarise routes as quickly and efficiently as OSPF.

Administrative Complexity and configuration overhead

Managing two protocols such as the RIP and OSPF is more complex than having only one protocol in place and these challenges.

Route redistribution

: This is the most critical and error-prone configuration task and the routes learned by OSPF on floor 2 must be then distributed into RIP on the business analyst router and there forth. The challenges of redistribution are as follows:

Administrative Distance Conflicts

The best practice for routers to choose the best path when there are multiple routing sources that provide information to the same network is Administrative Distance. OSPF is usually preferred over the RIP route but if a OSPF route is injected in RIP it will be seen as an external RIP route and if that same router also learns an OSPF route from the same network the OSPF route will be preferred which will cause troubleshooting problems which is another challenge of having two protocols in a network.

Troubleshooting

Troubleshooting for a more complex network which has more than one protocol is complex as network engineers do not use a single set of commands to get a complete view of the network. Diagnosing and finding the root problem between floor 1 and floor 3 will become substantially more complex as a network engineer needs more expertise as there are different protocol behaviours and their interaction.

Configuration Complexity:

Configuring these redistribution needs precise commands on border routers where the two protocols meet, and any mistake or misconfiguration is critical and lead to routing loops. It is important to ensure that maximum precision and accuracy is implemented when redistributing the RIP and OSPF between different floors.

Route Filtering

Without route filtering, redistribution of all routes will cause several problems such as symmetric routing paths which is caused if a route to the internet is learned by both protocols. This will cause admins to use route maps to be able to precisely find and control which routes are being shared between the protocols.

Impact on Failover and Backup Link Resilience

Backup links offer a good redundancy and fail over choice; it magnifies various challenges such as:

Unpredictable Convergence and routing loops

If the primary link between the floors fails, the OSPF domain will then converge and quickly use the backup link and redistribute the changed routes to the RIP domain. This would mean that the slow-converging RIP will still be advertising the now invalid primary path. During this time, the router in the RIP domain sends traffic back to the OSPF domain through the backup link to a destination that it thinks is still reachable via the primary link which will result in the creation of a transient routing loop (These are routing loops that are caused by the convergence of routing protocols due to topology changes)

Bad Path Selection

If the backup link between the shared router on floor 2 and 3 and the business analyst router this will result in a slower backup link connection. The OSPF would automatically assign this

slow link to an excessive cost and demote it to a backup role instead. In terms of the RIP, it only sees the hop count meaning that the primary path and backup path will both be 2 hops away as the RIP has no inherent way to prefer a higher bandwidth primary link over a lower bandwidth link. This will make them choose an unpredicted path which could lead to deficient performance.

In conclusion the hybrid network that Bright Soft has can be made to function, it will introduce significantly more challenges than a single network in terms of operational inefficiency (metric incompatibility and slow convergence). The administrative overhead that it brings in terms of the complex redistribution and complex troubleshooting and the compromised strength and toughness of the network (risk of transient loops). The backup link can also become a source of instability if not configured and implemented correctly. This all adds complexity to the hybrid network that Bright soft has implemented. For the longevity of the network it would be best recommended to migrate the entire network to a single Interior Gateway Protocol such as OSPF which will allow for the elimination of the complexities that were present with a dual/ hybrid protocol network. This will help build a more manageable network infrastructure that is easier to trouble shoot and is easily scalable.

(CCNA classes, 2022) (GeeksForGeeks, 2025)

References

Anon., 2025. GeeksForGeeks. [Online] Available at: <https://www.geeksforgeeks.org/computer-networks/difference-between-rip-and-ospf/> [Accessed 19 November 2025].

CCNA classes, 2022. ccna classes. [Online] Available at: <https://ccna-classes.com/ccna-exam-preparation/what-is-route-filtering-in-networking/> [Accessed 19 November 2025].

chiradeep-basumallick, 2024. spiceworks. [Online] Available at: <https://www.spiceworks.com/tech/networking/articles/access-control-lists/> [Accessed 2025 November 2025].

Fortinet, 2025. Fortinet. [Online] Available at: <https://www.fortinet.com/resources/cyberglossary/what-is-dmz> [Accessed 19 November 2025].

GeeksforGeeks, 2025. GeeksforGeeks. [Online] Available at: <https://www.geeksforgeeks.org/computer-networks/what-is-demilitarized-zone/> [Accessed 19 November 2025].

GeeksForGeeks, 2025. GeeksForGeeks. [Online] Available at: <https://www.geeksforgeeks.org/computer-networks/what-is-routing-loop-and-how-to-avoid-routing-loop/> [Accessed 19 November 2025].

PaloAlto, 2025. paloalto. [Online] Available at: <https://www.paloaltonetworks.com/cyberpedia/firewall-vs-ids-vs-ips> [Accessed 19 November 2025].

GeeksforGeeks (2022). DHCP Server Configuration in Cisco. [online] GeeksforGeeks. Available at: <https://www.geeksforgeeks.org/computer-networks/dhcp-server-configuration-in-cisco/>.

Anon, (2023). How to Configure DNS Server On Cisco Packet Tracer - Netizzan. [online] Available at: <https://netizzan.com/how-to-configure-dns-server-on-cisco-packet-tracer/>.

Tolga Bagci (2017). How to Configure Inter-VLAN Routing | SYSNETTECH Solutions. [online] SYSNETTECH Solutions. Available at: <https://www.sysnettechsolutions.com/en/configure-inter-vlan-routing-in-cisco-packet-tracer/>.

TECHNIG. (2015). How to Configure OSPF Routing Protocol Using Cisco Packet Tracer? [online] Available at: <https://www.technig.com/configure-ospf-routing-protocol/>.

GeeksforGeeks (2022). RIP Routing Configuration Using 3 Routers in Cisco Packet Tracer. [online] GeeksforGeeks. Available at: <https://www.geeksforgeeks.org/computer-networks/rip-routing-configuration-using-3-routers-in-cisco-packet-tracer/>.

ITExamAnswers (2021). 8.5.12 Packet Tracer - Configure Extended ACLs - Scenario 1 Answers. [online] ITExamAnswers.net. Available at: <https://itexamanswers.net/8-5-12-packet-tracer-configure-extended-acls-scenario-1-answers.html> [Accessed 24 Nov. 2025].

ccna tutorials (2023). Configure NAT in Packet Tracer: A Step-by-Step Guide - CCNA TUTORIALS. [online] CCNA TUTORIALS. Available at: <https://ccnatutorials.in/packet-tracer/configure-nat-in-packet-tracer-a-step-by-step-guide/>.